

Expressway Expansion, Export Competition, and Spatial Divergence*

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Abstract

We examine how domestic transportation infrastructure affects firm-level exports and spatial inequality by exploiting China's massive expressway expansion from 2000–2013. Using comprehensive customs data and precise firm-level measures of annual expressway proximity, we document sharp regional divergence: coastal firms experience 5.9% export increases while inland firms experience declines of 6.8–15.0%, creating a 21 percentage point coastal-inland difference. This divergence persists across all examined dimensions and is robust to instrumental variables approaches. We show these patterns reflect a competitive externality mechanism: when infrastructure reduces domestic costs, coastal firms with superior port access convert savings into larger delivered price reductions, expanding market shares and reducing residual demand for inland competitors. Export prices decline 5.4% coastally but increase 7.7% inland, and resource allocation improves coastally (+2.4%) but deteriorates inland (-3.6% to -4.7%). Infrastructure improvements that appear efficiency-enhancing on average can exacerbate spatial inequality through competitive reallocation when complementary inputs distribute unevenly across space.

JEL Classification: O18; F1; D2; R4

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1 Introduction

Large-scale transportation infrastructure investments are widely viewed as cornerstones of economic development. By lowering trade costs and improving market access, such investments are expected to integrate domestic markets, enhance firm productivity, and promote international trade participation. These expectations have motivated substantial public spending on transportation networks in developing economies over the past decades. Yet despite the scale of such investments, their consequences for firm-level export performance remain poorly understood. In particular, while evidence suggests infrastructure can reallocate economic activity spatially, whether road improvements generate broadly shared export gains across regions or systematically concentrate export activity in already-advantaged coastal locations—and through what mechanisms—remains unclear.

This question is particularly salient for export-oriented economies, where firm competitiveness depends critically on access to seaports—a geographic feature distributed highly unevenly across space (Clark et al., 2004). Improvements in domestic road infrastructure reduce transportation costs for firms throughout a country, but firms differ systematically in their access to ports. For coastal firms, reductions in road costs translate directly into lower total export costs and improved competitiveness in foreign markets. For inland firms, port-related frictions—longer transit times, coordination costs, and congestion—constitute a large share of total trade costs (Lima and Venables, 2001; Hummels and Schaur, 2013), such that equivalent reductions in road costs yield proportionally smaller reductions in overall export costs. Consequently, identical improvements in road connectivity may generate markedly different effects across space, potentially triggering competitive reallocation toward coastal regions rather than uniform gains (Krugman, 1991; Donaldson and Hornbeck, 2016).

We address this question by exploiting China’s massive expressway expansion as a natural experiment on exports from 2000 to 2013. During this period, China expanded its expressway network from approximately 16,000km to over 104,500km, coinciding with a dramatic surge in international trade. This setting offers three advantages for studying the infrastructure’s trade effects. First, the network’s staggered rollout across prefectures generates substantial spatial and temporal variation in firms’ exposure to infrastructure improvements. Second, detailed customs records at

the firm-product-country-year level permit precise measurement of export responses. Third, the stark geographic divide between coastal provinces with major seaports and inland provinces far from maritime infrastructure creates natural variation in firms' complementary access to international markets.

To guide the empirical analysis, we develop a tractable conceptual framework that distinguishes two components of trade costs: domestic transportation costs (road-related) and port-related costs (e.g., handling, storage, congestion, coordination). When expressway expansion reduces domestic costs across space, firms differ in translating these savings into international competitiveness based on their port access. Coastal firms with low port costs experience large proportional reductions in total trade costs, enabling delivered price declines that expand market shares. Inland firms with high port costs experience modest proportional reductions in total costs, limiting their ability to compete even when gaining road access. This asymmetry generates a competitive externality: as coastal firms expand production and capture export market shares, inland firms lose residual demand despite benefiting from the same infrastructure improvements.

The framework yields four testable predictions. Infrastructure improvements generate (1) average export increases that mask sharp coastal-inland divergence, with effects declining by distance from new connections; (2) gradual export responses emerging over multiple years as firms form buyer relationships and expand capacity; (3) opposing price movements, with coastal firms experiencing export price declines while inland firms may face price increases despite road improvements; and (4) market reallocation through entry-exit dynamics and resource flows, improving allocation coastally while worsening it inland. These predictions distinguish competitive externality from uniform cost reduction and guide our empirical analysis.

To implement the empirical analysis, we construct a novel firm-level exposure measure that captures annual changes in firms' proximity to the expressway network by combining three datasets. First, we manually digitize China's expressway network annually from 2000 to 2013 using transportation atlases, creating precise geospatial measures of network expansion. Second, we geocode firm locations and calculate each firm's distance to the nearest expressway in every year, generating a continuous, time-varying treatment measure. Third, we match these infrastructure exposures to customs records covering 35.7 million firm-product-country-year observations and industrial census data on 2.4 million firm-year observations. This combination enables us to exploit both spatial

variation (across firms) and temporal variation (within firms over time) in infrastructure exposure while controlling for unobserved firm-product and destination-year heterogeneity.

The results strongly confirm the framework’s predictions. First, using a difference-in-difference approach, we document sharp regional divergence in export responses. Firms in coastal Eastern China experience 5.9% export increases following expressway access. Firms in Central and Western inland China experience export declines of 6.8% and 15.0%, respectively, creating a 21 percentage point coastal-inland difference. These effects strengthen for firms closer to expressways (distance gradients), emerge gradually over 3–5 years (dynamic adjustment), and persist across alternative specifications, sample restrictions, and instrumental variables approaches using least-cost spanning tree networks and historical Ming Dynasty courier roads.

Second, heterogeneity analysis demonstrates the robustness of regional divergence. Across virtually all dimensions examined—manufacturing versus non-manufacturing, high versus low road dependence, state-owned versus private versus foreign ownership, OECD versus non-OECD destinations, capital versus intermediate versus final goods, and processing versus ordinary trade—the regional pattern persists: positive effects in Eastern China, negative effects in Central and Western China. This pervasive regional divergence rules out explanations based on industry composition, firm capabilities, or product characteristics, confirming that geographic position relative to ports determines infrastructure benefits.

Third, mechanism analysis confirms the competitive externality channel. Export prices decline 5.4% in Eastern China but increase 7.7% in Central China, directly validating the price pass-through prediction. Market reallocation operates through entry-exit dynamics and resource allocation shifts. Firm entry declines in Eastern China as incumbents consolidate, while Central and Western China experiences simultaneous increases in both entry and exit, indicating market churning. Resource allocation improves in Eastern China (misallocation decreases by 2.4%) but deteriorates inland (misallocation increases by 3.6–4.7%), revealing efficiency consequences of spatial reallocation. Together, these patterns confirm that infrastructure improvements trigger spatial reallocation through competitive dynamics rather than generating uniform cost savings.

These findings carry important policy implications. Standard infrastructure evaluations focusing on aggregate effects would conclude that China’s expressway expansion increased exports substantially. Our analysis reveals this average masks profound regional divergence: coastal re-

regions gained while inland regions lost, widening spatial inequality. The competitive externality mechanism demonstrates that policymakers cannot assume infrastructure benefits are distributed uniformly. Instead, improvements may amplify initial geographic advantages, concentrating activity in already-favored locations. Effective spatial development policy may require coordinated investments in multiple complementary inputs or explicit redistribution mechanisms to ensure benefits reach disadvantaged regions.

Our paper contributes to four literatures. First, we advance the literature on transportation infrastructure’s effects on international trade. Seminal work by [Lima and Venables \(2001\)](#) and [Hummels \(2007\)](#) establishes that transportation costs constitute a substantial barrier to trade, with infrastructure quality explaining significant variation in trade amounts across countries. Recent studies document the causal effects of specific infrastructure projects using natural experiments: [Donaldson \(2018\)](#) examines India’s colonial railway network; [Duranton et al. \(2014\)](#) studies the U.S. Interstate Highway System; [Baum-Snow et al. \(2017\)](#) analyzes China’s expressway expansion’s effects on urban form. These studies focus primarily on aggregate or regional impacts, leaving less understood how infrastructure shapes individual firms’ export decisions over time. The few firm-level studies that exist ([Martincus and Blyde, 2013](#); [Martincus et al., 2017](#)) make important progress by examining effects at one or two points in time, comparing pre- and post-infrastructure outcomes. We extend this work by providing firm-product-level evidence on the dynamic adjustment path of exports to road infrastructure, tracing how effects emerge gradually, accumulate over multiple years, and generate persistent spatial reallocation.

Second, we contribute to the growing literature examining China’s expressway network expansion. A pioneering study by [Faber \(2014\)](#) documents that peripheral regions lost economic activity to core areas following expressway connections. Subsequent work confirms the effects on GDP ([He et al., 2020](#)), firm productivity ([Yang, 2018](#)), market integration ([Fan et al., 2023](#)), firm markups ([Wu et al., 2023](#)), and urban spatial structure ([Baum-Snow et al., 2017](#)). While these studies consistently document spatial heterogeneity, they focus primarily on domestic outcomes and typically rely on binary treatment definitions at administrative-unit levels.

We advance this literature in two complementary ways. First, methodologically, we advance infrastructure evaluation methodology by digitizing China’s expressway network expansion from 1992 to 2013 and constructed continuous, time-varying treatment measures by calculating each

firm’s annual distance to the nearest expressway from 2000 to 2013. Our approach tracks proximity changes year by year, generating distance gradient and dynamic treatment indicators that vary both across and within firms over time. This firm-level measurement reclassifies 32% of observations relative to county-level definitions and enables distance-gradient specifications that directly test spatial decay patterns. While recent studies use distance-based measures to examine infrastructure effects on productivity ([Holl, 2016](#)) and manufacturing location ([Ghani et al., 2016](#)), our time-varying construction exploits staggered network expansion timing through difference-in-differences and event study designs, providing stronger causal identification of dynamic treatment effects. Second, substantively, we document a novel empirical pattern: export effects depend critically on firms’ proximity to seaports. The interaction between domestic road connectivity and port access generates sharp coastal-inland divergence—a 21 percentage point gap that persists across all examined dimensions—offering a parsimonious explanation for the spatial heterogeneity documented in prior work.

Third, we contribute to the literature on infrastructure’s role in spatial development and inequality. A longstanding question concerns whether infrastructure reduces or exacerbates spatial inequality. Studies of within-metropolitan areas often find that highways reduce inequality by enabling suburban development ([Baum-Snow, 2007, 2020](#)), while research on inter-regional infrastructure in developing countries documents concentration effects with peripheral regions losing activity ([Faber, 2014; He et al., 2020](#)). We reconcile these seemingly contradictory findings by showing they depend on whether improved connectivity links firms to larger external markets. In our setting, expressway expansion benefits coastal firms with superior port access while harming inland firms distant from maritime infrastructure. More broadly, we demonstrate that infrastructure’s spatial effects depend critically on the geographic distribution of complementary inputs. This insight extends recent theoretical work on spatial sorting ([Redding and Turner, 2015](#)) and place-based policies ([Kline and Moretti, 2014](#)) by showing that infrastructure effects are fundamentally shaped by interactions with pre-existing geographic advantages.

Fourth, we identify the mechanisms through which infrastructure generates spatial reallocation. Recent work emphasizes the importance of understanding not just whether infrastructure affects outcomes, but how ([Donaldson, 2018](#)). We show that infrastructure-induced reallocation operates through competitive externality via three complementary channels: export price pass-through

(coastal prices decline 5.4% while inland prices increase 7.7%), entry-exit dynamics (coastal entry declines as incumbents consolidate while inland markets experience churning), and resource allocation shifts (allocation improves 2.4% coastally but deteriorates 3.6–4.7% inland). These findings advance understanding of competitive externalities in trade. While prior work documents that trade liberalization can generate winners and losers through competitive reallocation (Melitz, 2003; Bernard et al., 2007), we show that domestic infrastructure improvements can produce similar effects when regions differ in access to export markets. The price divergence—coastal declines, inland increases—provides direct evidence for competitive externality versus mechanical cost reduction.

The remainder of the paper proceeds as follows. Section 2 provides background on China’s expressway expansion. Section 3 develops the conceptual framework and derives testable predictions. Section 4 describes the data and empirical strategy. Section 5 presents empirical results including average effects, dynamics, heterogeneity, and instrumental variables. Section 6 investigates mechanisms through price pass-through and market reallocation. Section 7 concludes.

2 Institutional Background: The National Expressway Network

The Chinese road system comprises national, provincial, county, and township roads, as depicted in Figure A2 in Appendix A. National roads include toll-charging expressways and toll-free highways. National expressways consist of national trunk expressways, which connect different provinces, as well as local short expressways that provide access to these trunk routes. Although national highways and provincial roads also form a national-level road network, they are generally narrower, slower, in poorer condition, and less convenient than expressways. Since our study focuses on inter-regional cargo transportation for international exports, the term “national expressway network” in this paper refers specifically to the national trunk expressway network.

Figure 1 illustrates the evolution of China’s national expressway network from 1992 to 2013. Over this period, expressway construction proceeded in three broad stages. The first stage, spanning the late 1970s to the early 1990s, marked the initial adoption of expressways in China. At the outset of economic reform, long-distance road transportation was limited, and expressways were largely absent. After extensive policy debate, the central government initiated construction of

China's first expressway in 1984, connecting Shenyang and Dalian—two major cities in northern China. The project was completed in 1990, with a total length of approximately 375 km.

The second stage, from the early 1990s to 2004, saw the systematic planning and rapid expansion of the expressway network. During this period, the government introduced the “5–7” National Trunk Expressway Plan in 1992, which aimed to construct a network exceeding 35,000 km and connect all provincial capitals and cities with populations above 500,000. The plan consisted of five north–south and seven east–west expressways, covering more than 600 million people—approximately half of China's population at the time. Although originally scheduled for completion by 2020, expansion accelerated substantially following the 1997 Asian Financial Crisis as the government used infrastructure investment to stimulate economic growth. The network was completed by 2007, more than a decade ahead of schedule, with an average annual construction of roughly 4,000 km.

The third stage began after 2004, when China further expanded its expressway ambitions through the “7–9–18” network plan. This plan envisioned seven radial expressways from Beijing, nine north–south corridors, and eighteen east–west corridors, totaling more than 85,000km. The expanded network aimed to connect major transportation hubs, seaports, tourist destinations, and cities with populations exceeding 200,000, ultimately serving over one billion people.¹ Although originally projected to span three decades, the large-scale fiscal stimulus implemented in response to the 2008 global financial crisis substantially accelerated construction. By 2013, the expressway network was nearly complete—approximately 20 years ahead of schedule. As shown in Figure 1, over our sample period, total expressway mileage increased from about 16,000km in 2000 to more than 104,500km in 2013.

The rapid expansion of the expressway network dramatically reduced transportation time and costs, providing a faster, safer, and more reliable alternative to traditional roads. Using micro-level cost data, [Fan et al. \(2023\)](#) estimate that transportation costs on expressways were approximately 20% lower than on regular roads between 2000 and 2010. In contrast, long-distance transport via non-expressway roads often involves multiple tolls, inspection fees, and informal charges, as well as poor road conditions and higher risks of cargo loss. Rail transport, while safer, has

¹The network directly covers more than 90% of the eastern region, 83% of the central region, and nearly 70% of the western region, accounting for over 85% of the national GDP.

limited geographic coverage and lacks door-to-door connectivity, making it less suitable for firm-level freight shipments, particularly for export-oriented manufacturing firms requiring access to seaports.

Importantly for identification, expressway construction was highly uneven across space and time. Coastal provinces—particularly in the northeast and east—were connected earlier and more densely, while central and western regions experienced substantial delays in network access. This geographic pattern reflected both topographical constraints and proximity to major seaports rather than pre-existing export trends. Financing arrangements also varied, with approximately 15% of funding provided by the central government, 70–80% financed through domestic bank loans and local government borrowing, and a small share from private investors. Construction contracts were predominantly awarded to state-owned enterprises through centralized planning processes. These features generated substantial variation in firms’ exposure to the expressway network over time that is plausibly exogenous to individual firm-level export outcomes, conditional on our fixed effects specifications.

Over the past three decades, expressways have become the dominant mode of long-distance freight transportation in China. As shown in Figure A3 in Appendix A, road transport accounts for the majority of cargo movements and has expanded more rapidly than rail or waterway transport. Although air transportation has also grown, it primarily serves passenger travel and plays a limited role in freight logistics. In the empirical analysis, we further address potential confounding from other transportation modes through a series of robustness checks.

3 Conceptual Framework

Our theoretical framework builds on the literature on trade costs, market access, and spatial reallocation in economic geography and international trade (Redding and Turner, 2015; Donaldson and Hornbeck, 2016; Donaldson, 2018). We extend this perspective to firm-level export markets by explicitly distinguishing domestic road connectivity from access to seaports, thereby generating predictions about heterogeneous infrastructure effects across space.

This distinction is motivated by a simple observation: expressway improvements reduce domestic transportation costs, but their impact on total export costs—and thus on competitiveness—depends

on the relative importance of port-related frictions in firms' cost structures (Lima and Venables, 2001; Clark et al., 2004; Hummels and Schaur, 2013). Port-related frictions include handling, storage, congestion, coordination with shipping agents, and administrative procedures—costs that are higher for inland firms due to longer transit times, greater reliance on intermediaries, and limited direct port access. For inland firms with high port-related costs, domestic road improvements constitute a smaller share of total cost reductions; for coastal firms with low port-related costs, the same road improvements translate into larger proportional gains.² This asymmetry provides a parsimonious framework for interpreting the spatial heterogeneity we document empirically.

3.1 Setup

Consider an economy with multiple regions $r \in \{C, I\}$, where C denotes coastal regions with direct access to seaports and I denotes inland regions that are geographically distant from ports. Firms indexed by i are heterogeneous in productivity ϕ_i , drawn from a distribution $G(\phi)$ with support $[\underline{\phi}, \infty)$, and are located in region $r(i)$. Firms produce differentiated varieties under monopolistic competition and may export to foreign destination markets d .

Production uses labor as the sole input. The marginal production cost of firm i is given by

$$mc_i = \frac{w_{r(i)}}{\phi_i} \quad (1)$$

where $w_{r(i)}$ denotes the regional wage rate. In the baseline model, regional wages are treated as exogenous; Section 3.5 discusses how endogenous factor mobility amplifies the regional divergence in export prices.

Exporting involves trade costs that arise along the domestic and international transportation chain. We distinguish three components. The first, τ_i^{road} , captures domestic transportation costs from the factory to the export port. The second, $\tau_{r(i)}^{port}$, reflects access-to-seaport costs, with inland firms facing persistently higher $\tau_{r(i)}^{port}$ than coastal firms. The third, τ_d^{sea} , represents international shipping costs from the port to destination d . We model all three components as iceberg trade

²To illustrate the mechanism, suppose coastal and inland firms initially face total trade costs of $\tau^{coastal} = 1.2$ and $\tau^{inland} = 1.5$. Expressway expansion reduces domestic costs by 20% for both. For coastal firms ($\tau_r^{port} = 1.1$), total costs decline from 1.2 to 1.08 (a 10% reduction). For inland firms ($\tau_r^{port} = 1.3$), total costs decline from 1.5 to 1.46 (a 2.7% reduction). This 7.3 percentage-point difference in cost reductions translates directly into divergent export competitiveness through the demand elasticity.

costs, so the total trade cost for firm i exporting to destination d is

$$\tau_{ird} = \tau_i^{road} \cdot \tau_{r(i)}^{port} \cdot \tau_d^{sea} \quad (2)$$

Expressway expansion directly reduces τ_i^{road} , while $\tau_{r(i)}^{port}$ reflects persistent geographic and institutional frictions that vary systematically across regions and evolve slowly over time.

3.2 Export Prices

Firms compete monopolistically in destination market d with constant elasticity of substitution $\sigma > 1$. Prices are set as constant markups over marginal cost. We measure export prices as free-on-board (FOB) prices, which include domestic transportation costs from the factory to the port and port-related pre-shipment charges. The factory-gate price is

$$p_i^{XW} = \mu \cdot mc_i = \mu \cdot \frac{w_{r(i)}}{\phi_i} \quad (3)$$

where $\mu = \sigma / (\sigma - 1)$ is the constant markup. The observed FOB price is

$$p_{ir}^{FOB} = p_i^{XW} \cdot \tau_i^{road} \cdot \tau_{r(i)}^{port} \quad (4)$$

and the delivered price in destination d is

$$p_{ird}^{del} = p_{ir}^{FOB} \cdot \tau_d^{sea} \quad (5)$$

FOB prices serve as an intermediate outcome linking infrastructure improvements to export competitiveness. A reduction in τ_i^{road} lowers p_{ir}^{FOB} and hence p_{ird}^{del} , enabling the firm to expand market share in destination d .

3.3 Export Quantities and the Entry Decision

Demand. Destination-country consumers have CES preferences over differentiated varieties. The destination-country CES price index is

$$P_d^{1-\sigma} = \sum_{i \in \Omega_d} \left(\frac{\mu w_{r(i)}}{\phi_i} \cdot \tau_i^{road} \cdot \tau_{r(i)}^{port} \cdot \tau_d^{sea} \right)^{1-\sigma} \quad (6)$$

where Ω_d is the endogenous set of exporters to d . Setting $A_d = Y_d \cdot P_d^{\sigma-1}$, with Y_d total destination expenditure, firm i 's export quantity to destination d is

$$Q_{ird} = A_d \left(p_{ird}^{del} \right)^{-\sigma} = A_d \left(\frac{\mu w_{r(i)}}{\phi_i} \cdot \tau_i^{road} \cdot \tau_{r(i)}^{port} \cdot \tau_d^{sea} \right)^{-\sigma} \quad (7)$$

The partial-equilibrium response of export quantities to an expressway improvement, holding P_d fixed, is

$$\Delta \ln Q_{ird} \big|_{\text{partial}} = -\sigma \cdot \Delta \ln \tau_i^{road} \quad (8)$$

This elasticity is identical for all firms given the same $\Delta \ln \tau_i^{road}$. However, as shown in Section 3.4, the general-equilibrium response diverges sharply between coastal and inland firms once P_d adjusts endogenously.

Export participation. Exporting requires payment of a fixed cost f_x (denominated in units of labor), so the entry cost is $w_{r(i)} \cdot f_x$. Firm i participates in export market d if and only if variable export profits cover this fixed cost:

$$\pi_{ird} = \frac{1}{\sigma} \left(\frac{p_{ird}^{del}}{P_d} \right)^{1-\sigma} Y_d \geq w_{r(i)} \cdot f_x \quad (9)$$

Rearranging yields a productivity cut-off $\phi_{r,d}^*$ below which firm i does not export:

$$\phi_{r,d}^* = \left(\frac{\sigma w_{r(i)}^\sigma f_x (\tau_i^{road} \cdot \tau_{r(i)}^{port} \cdot \tau_d^{sea})^{\sigma-1}}{Y_d / P_d^{1-\sigma}} \right)^{\frac{1}{\sigma-1}} \quad (10)$$

Three properties of equation (10) are central to the analysis. First, $\phi_{r,d}^*$ is increasing in $\tau_{r(i)}^{port}$: inland firms face a higher productivity bar because larger port frictions raise total export costs and lower operating profits, so fewer inland firms export in equilibrium. Second, expressway expansion lowers τ_i^{road} and reduces $\phi_{r,d}^*$; the absolute reduction is larger for coastal firms, where road improvements constitute a larger share of total trade costs. Third, and most importantly for entry and exit patterns: when coastal expansion lowers P_d (Section 3.4), the profitability condition (equation (9)) tightens for inland firms, potentially raising $\phi_{I,d}^*$ even as τ_i^{road} falls, which can generate simultaneous increases in both entry into and exit from exporting among inland firms.³

3.4 Market Competition and Spatial Reallocation

The partial-equilibrium responses above represent only part of the adjustment process, as expressway expansion also induces reallocation through intensified competition in export markets.

The key mechanism operates through the destination-country price index P_d (equation (6)). When expressway expansion reduces coastal firms' τ_i^{road} , the coastal term in $P_d^{1-\sigma}$ increases since $1 - \sigma < 0$. Substituting into equation (7), the general-equilibrium response of an inland firm's exports is

$$\Delta \ln Q_{ird} \big|_{i \in I} = \underbrace{-\sigma \cdot \Delta \ln \tau_i^{road}}_{\text{direct cost effect (+)}} + \underbrace{(\sigma - 1) \cdot \Delta \ln P_d}_{\text{competitive externality (-)}} \quad (11)$$

where the second term captures the loss of residual demand as P_d falls due to coastal expansion. Inland exports decline when the competitive externality dominates:

$$(\sigma - 1) \cdot |\Delta \ln P_d| > \sigma \cdot |\Delta \ln \tau_i^{road}| \quad (12)$$

This condition is more likely to hold when: (i) coastal firms hold a large pre-existing share of destination market d , so that their expansion generates a large decline in P_d ; (ii) the demand elasticity σ is large, amplifying both the competitive externality and the price-competitiveness channel; and (iii) port access is highly asymmetric, so that inland firms' direct cost savings $|\Delta \ln \tau_i^{road}|$ are small relative to coastal firms'. In our setting, 91% of total Chinese exports originate from Eastern

³where “entry” and “exit” refer to changes in export market participation rather than firm creation or dissolution.

China, satisfying condition (i), and the literature (Broda and Weinstein, 2006) estimates $\sigma \in [3, 8]$ for manufactured goods, satisfying condition (ii). Together, these features make the competitive externality mechanism quantitatively plausible, and the empirical analysis confirms that inland exports decline despite their own cost reductions.

The market share of firm i in destination d makes the competitive interdependence explicit:

$$s_{ird} = \frac{(\phi_i / w_{r(i)} \tau_i^{road} \tau_{r(i)}^{port} \tau_d^{sea})^\sigma}{\sum_{j \in \Omega_d} (\phi_j / w_{r(j)} \tau_j^{road} \tau_{r(j)}^{port} \tau_d^{sea})^\sigma} \quad (13)$$

When coastal firms' τ_i^{road} falls, coastal market shares expand, compressing inland residual demand. This effect is stronger when coastal productivity is more dispersed (concentrating gains among top exporters) and when σ is larger.

3.5 Factor Mobility and Inland Price Amplification

The baseline model treats regional wages as exogenous. We now extend the framework to show how endogenous factor mobility reinforces regional divergence in export prices, providing a theoretical basis for the inland FOB price increases documented in Section 6.

Suppose regional labor supply is upward-sloping: $w_r = \bar{w}_r \cdot L_r^\gamma$ for $\gamma > 0$, where L_r is total regional employment. As coastal exporters expand production and inland market shares contract, labor reallocates toward coastal regions, generating

$$\Delta L_C > 0 \implies \Delta w_C > 0 \quad (\text{coastal wages rise}), \quad (14)$$

$$\Delta L_I < 0 \implies \Delta w_I < 0 \quad (\text{inland wages fall as workers migrate out}), \quad (15)$$

with the net effect on inland FOB prices given by

$$\Delta \ln p_{ir}^{FOB} \big|_{i \in I} = \underbrace{\Delta \ln \tau_i^{road}}_{\text{road cost saving } (-)} + \underbrace{\Delta \ln w_I}_{\text{wage change } (-)} - \underbrace{\Delta \ln \phi_i}_{\text{composition effect } (+)} \quad (16)$$

Although both $\Delta \ln \tau_i^{road} < 0$ and $\Delta \ln w_I < 0$ may push prices down, the composition effect $-\Delta \ln \phi_i > 0$ pushes prices up: the departure of more productive workers and capital toward coastal regions

shifts the productivity distribution of surviving inland exporters downward, raising effective unit costs for remaining firms. Additionally, the clustering of firms near new expressway access points intensifies local input-market competition in inland regions where factor markets are thin, generating local factor price pressures that can dominate the direct road cost savings. When these forces dominate the cost savings, inland FOB prices rise net—precisely the pattern we document empirically (Section 6).

We treat this extension as a complementary amplification mechanism rather than a core prediction of the framework, as it requires additional assumptions about factor market structure that we do not test directly. It nonetheless provides a theoretically grounded interpretation for an otherwise surprising empirical regularity.

3.6 Theoretical Predictions

The framework organizes our empirical analysis four testable predictions.

Prediction 1. *(Average effect and regional heterogeneity) Expressway expansion increases average exports, but this average masks sharp regional divergence. Coastal firms experience export gains as road cost savings translate into lower delivered prices and expanded market shares. Inland firms may experience export declines despite their own cost reductions: coastal expansion lowers P_d , eroding residual demand through the competitive externality in equation (11). Within each region, effects vary with distance from the expressway, but in opposite directions—gains deepen for coastal firms closer to the network, while losses deepen for inland firms closer to the network.*

Prediction 2. *(Gradual dynamic adjustment) Export responses materialize gradually over a multi-year horizon rather than immediately, reflecting adjustment frictions in forming buyer–supplier relationships, expanding logistics capacity, and learning about foreign market opportunities (Albornoz et al., 2012; Eaton et al., 2025). Effects accumulate over time rather than arriving as a one-time level shift.*

Prediction 3. *(Asymmetric export price pass-through) Expressway expansion generates opposing FOB price movements across regions. Coastal firms pass road cost savings into lower FOB prices, improving competitiveness and driving quantity gains. Inland firms may face rising FOB prices*

despite road cost reductions, as competitive pressures from coastal expansion and factor outflows dominate direct cost savings. This divergent price pass-through provides the key empirical signature distinguishing the competitive externality mechanism from a uniform cost-reduction story.

Prediction 4. *(Market reallocation and concentration) Expressway expansion generates opposing structural effects across regions. In coastal regions, established exporters expand their export portfolios, resource allocation improves as factors flow toward high-productivity exporters, and entry may decline as competitive thresholds rise. In inland regions, export market participation churns with simultaneous increases in entry and exit, and resource misallocation worsens as factors flow outward. Market concentration declines as expressway access enables broader firm participation in export markets.*

The model has ambiguous aggregate welfare implications. Efficiency gains from reallocation toward high-productivity coastal firms are offset by spatial redistribution that generates clear regional winners and losers. A full welfare analysis would require modeling domestic product markets, factor mobility, and regional price indices—extensions beyond the scope of this paper.

3.7 Mapping to the Empirical Design

The theoretical framework maps directly into the empirical analysis. Export quantities Q_{irdt} constitute the primary outcome, measured using firm-product-country-year customs data. Firms' exposure to expressway expansion proxies changes in τ_i^{road} , measured through time-varying distance from each firm to the nearest expressway segment. Differences in $\tau_{r(i)}^{port}$ generate heterogeneous responses across coastal and inland regions, which we exploit through sample splits and interaction specifications. The empirical analysis tests Prediction 1 through distance-gradient and difference-in-differences specifications, with identification supported by two instrumental variables (Section 5); Prediction 2 through event-study designs validated with the [De Chaisemartin and d'Haultfoeuille \(2020\)](#) estimator and [Rambachan and Roth \(2023\)](#) honest DID approach (Section 5); and Predictions 3 and 4 through a suite of mechanism tests including export price regressions, entry–exit analysis, market concentration, and resource misallocation estimates (Section 6).

4 Data and Empirical Method

4.1 Data

Our analysis uses annual data from 2000 to 2013 and combines firm-level trade and production data with detailed geographic information on China’s expressway network. The empirical analysis draws on multiple complementary datasets, which together allow us to construct firm-level exposure to expressway expansion, measure export outcomes, and implement instrumental-variable strategies.

Annual Dynamic Expressway Network Data. A central component of our analysis is a newly constructed annual expressway network covering the period 1992–2016. Constructing time-varying expressway shapefiles is nontrivial, as no official digital dataset reports opening years at the link level. We therefore combine two complementary sources: (i) the 2016 expressway shapefile from the Chinese Geographical Information Monitoring Cloud Platform, which reports expressway names and the counties they traverse, and (ii) annual printed transportation atlases published by SinoMaps Press and other major publishers, as detailed in Table B1 in Appendix B.⁴

To ensure consistency and accuracy, we restrict attention to atlases published at the beginning of each year and prioritize volumes from the same publisher. These atlases document when expressways reached specific county-level units. Using this information, we manually digitize expressway openings year by year and reconstruct a national expressway network for each year between 1992 and 2016. Since our focus is on long-distance freight transport, we retain only national expressways and exclude feeder roads and urban ring roads.

Chinese Customs Trade Data. Firm-level export outcomes are obtained from the Chinese Customs Trade Statistics (CCTS) compiled by the General Administration of Customs of China. This dataset covers the universe of Chinese trade transactions and reports annual export activity at the firm-product-country-year level. The data include exports to 243 destination countries across 7,526 products classified under the eight-digit Harmonized System (HS) for all industries. The dataset provides detailed transaction-level information, including product characteristics, quantity, FOB

⁴Data from SinoMaps Press begins in 1992, as expressways were virtually nonexistent in China before 1992.

price, revenue, measurement units, and firm identifiers such as firm name, trade nature, contact information, and location. For the purposes of our analysis, we aggregate the data to the firm—product—country—year level and focus exclusively on export transactions. Firms with multiple establishments are treated as separate entities.

Chinese Industry Firm Data. To examine mechanisms underlying export responses, we use the Chinese Annual Survey of Industrial Firms (CASIF) compiled by the National Bureau of Statistics. It contains annual data covering all state-owned enterprises (SOEs) and non-SOEs with annual sales of at least 5 million CNY (approximately \$785,053) before 2007, all firms with annual sales of at least 5 million CNY between 2007 and 2010, and all firms with annual sales of at least 20 million CNY (approximately \$3.08 million) since 2011. Additionally, the CASIF database contains detailed information about manufacturing firms, including their identification (e.g., firm name, firm nature, setup year, setup month, telephone number, zip code, and contact person), output, employment, assets, and value-added. Like the U.S. North American Industry Classification System (NAICS), the CASIF classifies each firm into an industry using a four-digit Chinese Industry Classification System. We follow standard data-cleaning procedures to address reporting errors and missing observations; details are provided in Appendix C.

Two Geocoding Datasets. To link firms to the expressway network, we geocode plant locations using Baidu Maps. The CCTS and CASIF datasets provide detailed plant names and addresses, which we use to obtain geographic coordinates through Baidu’s geocoding API.⁵ We geocode each firm’s location separately using both the text address and firm name fields to cross-validate coordinates. Baidu Maps returns latitude-longitude coordinates along with a reliability score for each geocoding attempt. We retain observations meeting two quality thresholds: (1) geocoding reliability of at least 25%, corresponding to a spatial error below 8 kilometers, and (2) coordinate discrepancy between the two geocoding methods (address-based and name-based) of less than 1 kilometer.⁶ This procedure yields geocoded coordinates for 90% of CCTS observations and 92%

⁵While more than half of the addresses in the CASIF dataset are invalid, 94% of the observations include a 12-digit address code that accurately represents the firm’s village or neighborhood committee. Given China’s high population density and the compact geographic area of these local units, we use Python to extract addresses based on the 12-digit codes, treating these as firm addresses where original address data are missing.

⁶Discrepancies between geocoding methods may arise from measurement error or firm relocations between survey waves. Both sources of variation are orthogonal to expressway placement timing and therefore introduce classical

of CASIF observations.

Three Datasets for Instrumental Variables. To address potential endogeneity in expressway placement, we construct two instrumental variables using an additional three datasets. The first exploits historical transportation routes from the Ming Dynasty (1368-1644) Courier Road shapefile in 1587, which is sourced from the Harvard Dataverse ([Berman and Zhang, 2017](#)).⁷ The courier road network was originally developed for military purposes and connected major cities during the Ming Dynasty, significantly promoting economic development and facilitating trade between regions.

The second instrumental variable is constructed using two additional datasets: land cover raster data from the Scientific Data Center of the Chinese Academy of Sciences and elevation raster data from the China Resources Satellite Application Center. Both datasets are from 2000 and have a spatial resolution of 30×30 meters. Using these data, we construct a least-cost path expressway network that captures exogenous variation in construction feasibility and serves as an instrument for the 2013 expressway network. The construction cost map is shown in Figure A7 in Appendix A. Our approach builds on the least-cost network methodology commonly used in the infrastructure literature (e.g., [Faber, 2014](#)), but is tailored to a firm-level setting with time-varying exposure.

The remaining data used in our paper are sourced from the Chinese Statistical Yearbook and the Chinese Transportation Yearbook. For all datasets mentioned above, we deflate the monetary variables to the year 2000 using two-digit HS product-level price indices and two-digit industry deflators provided by the Chinese National Bureau of Statistics.⁸

Since intermediary firms do not engage in production but only facilitate product delivery, accurately determining the distance between production sites and the expressway network is challenging, which may introduce bias into the main results. To address this issue, we create a sample that excludes intermediary firms, enabling a more precise calculation of the distance between each production plant and the expressway network.⁹

measurement error that attenuates our estimates toward zero.

⁷Data source: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/SB8ZTM>.

⁸The deflated variables include export prices, total assets, fixed assets, currency assets, profits, subsidies, production costs, and intermediate inputs.

⁹Firms are classified as intermediaries if their names contain terms such as ‘jing mao,’ ‘shang mao,’ ‘mao yi,’ ‘jin chu kou,’ ‘huo yun,’ or ‘wu liu.’

We combine the geocoded firm locations with the annual expressway network data to construct two complementary analysis samples. To ensure a comparable treatment and control group, we restrict both samples to firms located within 100km of the expressway network at some point during the sample period. This restriction excludes firms in remote areas that remain far from the network throughout 2000–2013 and would not contribute to identification. Sample A combines expressway exposure with customs data and is used to study export outcomes. Sample B links expressway exposure to industrial firm data and is used to analyze mechanisms. In both samples, treatment status is defined by firms’ time-varying distance to the nearest expressway.

Table 1 presents summary statistics for Sample A, our primary dataset for export analysis.¹⁰ The sample contains 35.7 million firm-product-country-year observations from 2000–2013, covering Chinese manufacturing exporters’ transactions at the eight-digit HS product level. Firms are geographically concentrated near the expressway network, with 91% of total exports originating from Eastern China—the coastal region closest to major international ports (Figure A4 in Appendix A). Foreign-owned and privately-owned firms dominate the sample, together accounting for over 95% of observations. The table also shows that expressway expansion substantially reduces firms’ distances to the network, with the largest reductions among private firms (50% decline) and firms in Eastern and Central China.

Figure 2 reveals that exporters concentrate in coastal cities and produce substantially more product varieties than inland firms, consistent with coastal firms’ superior market access. Over the study period, private firms and their exports experienced substantial growth, coinciding with the expressway network’s expansion.

4.2 Empirical Strategy

Measuring Firm-Level Exposure to Expressways. The staggered expansion of China’s expressway network generates variation in firms’ access to transportation infrastructure that differs across space and over time. We exploit this variation to identify the causal effects of expressway access on firm-level export outcomes using a difference-in-differences (DID) framework. Our empirical strategy proceeds in three steps. First, we estimate distance gradients showing how export

¹⁰Sample B, used for productivity, market concentration, and misallocation analysis, contains 2.4 million firm-year observations and is described in Table B2 in Appendix B.

effects vary with proximity to the network. Second, we collapse these distance bands into a binary treatment indicator to estimate average effects. Third, we employ event-study designs to examine pre-trends and trace dynamic adjustment paths.

Treatment is defined at the firm level and varies dynamically over time. Using annually digitized expressway shapefiles and geocoded firm locations, we compute each firm's distance to the nearest expressway segment in every year. Firms are considered exposed to the expressway network when their distance to the nearest expressway falls below a given threshold. This design allows firms to transition from control to treatment status as the network expands, enabling within-firm comparisons of export outcomes before and after gaining expressway access. For example, a firm located 50km from the nearest expressway in 2003 may find itself 15km away by 2007 as new segments open nearby, transitioning from control to treatment and contributing to identification through its own temporal variation.

To flexibly capture how export responses vary with proximity to the expressway network, we partition firms into 10-kilometer distance bands up to 60 kilometers, with firms located 60–100km serving as the control group. We choose 10-kilometer bins to balance granularity with statistical precision.

Formally, our specification is given by:

$$\ln(Q_{igdt}) = \beta_0 + \sum_{r \in R} \beta_r * \mathbb{1}\{Distance_{irt}\} + \lambda_{ig} + \gamma_{dt} + \varepsilon_{igdt} \quad (17)$$

where $\ln(Q_{igdt})$ denotes the logarithm of export amounts of product g exported by firm i to destination country d in year t . The indicator $\mathbb{1}\{Distance_{irt}\}$ equals one if firm i is located within distance range $r \in \{0-10, 10-20, 20-30, 30-40, 40-50, 50-60\}$ kilometers of the expressway network in year t . Firms located 60–100 kilometers away constitute the omitted reference group.

The coefficients β_r capture the semi-elasticity of exports with respect to proximity to the expressway network relative to firms that remain distant from the network. The specification includes firm-6-digit product fixed effects (λ_{ig}), which absorb all time-invariant characteristics of a given firm-product pair, including location, industry affiliation, and baseline productivity. Destination-year fixed effects (γ_{dt}) control for time-varying conditions in export markets, such as destination country demand, exchange rates, trade policy, and other bilateral shocks common to all firms exporting to

country d in year t . Together, these fixed effects ensure that identification comes from within-firm-product variation in expressway proximity over time, conditional on destination-specific annual shocks. Standard errors are clustered at the firm level to account for serial correlation within firms and cross-product dependence in export responses, consistent with the firm-level variation in expressway exposure. Results are robust to alternative clustering at the county-industry and county-year levels. Intuitively, identification comes from comparing the same firm-product pair's exports to the same destination before and after the firm gains expressway access, differencing out time-invariant firm-product characteristics and destination-specific temporal shocks.

Baseline Difference-in-Differences Design. While equation (17) flexibly captures distance gradients, our baseline specification consolidates these effects into a single treatment indicator for parsimony and to facilitate heterogeneity analysis across regions, industries, and firm types.

$$\ln(Q_{igdt}) = \beta_0 + \beta_1 * Treat_{it} + \lambda_{ig} + \gamma_{dt} + \varepsilon_{igdt} \quad (18)$$

where $Treat_{it}$ is a binary indicator equal to one if firm i is located within 40km of the expressway network in year t , and zero otherwise. Firms located 40–100 kilometers serve as the control group. We set the treatment threshold at 40km based on the distance gradient estimates in equation (17), which show statistically significant and economically meaningful effects within this range while effects beyond 40km are indistinguishable from zero. This data-driven cutoff captures firms' effective access to the expressway network while maintaining a clear distinction between treated and control groups. The coefficient β_1 captures the average effect of expressway access on firm-level exports. All other variables are defined as in equation (17).

Event-Study Design. To examine the dynamic effects of expressway expansion and assess the validity of the parallel trends assumption, we employ an event-study design that traces firms' export responses before and after gaining access to the expressway network.

$$\ln(Q_{igdt}) = \beta_0 + \sum_{k=-6}^{k=-2} \beta_k * \mathbb{1}\{After_{it} = k\} + \sum_{k=0}^{k=7} \beta_k * \mathbb{1}\{After_{it} = k\} + \lambda_{ig} + \gamma_{dt} + \varepsilon_{igdt} \quad (19)$$

where $After_{it}$ denotes the number of years relative to the first year in which firm i becomes located within 40km of the expressway network. The indicator $\mathbb{1}\{After_{it} = k\}$ equals one if year t is k years away from the firm’s initial expressway exposure. The year immediately preceding treatment ($k = -1$) is omitted and serves as the reference period. For example, if a firm first becomes exposed to the expressway network in 2005, then $k = 0$ in 2005, $k = 1$ in 2006, and $k = -1$ in 2004. To limit the influence of sparsely populated tails, event time is top- and bottom-coded at $k = -6$ and $k = 7$, respectively. All other variables are defined as in equation (17).

The event-study coefficients β_k trace the evolution of treatment effects relative to the year immediately preceding expressway access. If expressway expansion has immediate effects, we expect β_0 to differ from zero; if effects emerge gradually, we expect β_k to grow over time in the post-treatment period. Pre-treatment coefficients provide a test of the parallel trends assumption. Under this assumption, coefficients on pre-treatment indicators ($k < 0$) should be statistically indistinguishable from zero, while post-treatment coefficients ($k \geq 0$) trace the dynamic causal effects of expressway access on exports.

5 Empirical Findings

This section tests the first two predictions derived from the conceptual framework, documenting three main findings. First, expressway expansion increases exports by 5.9% on average, with effects declining steeply by distance—but this average masks a 21 percentage point coastal-inland difference (Section 5.1). Second, effects emerge gradually over 3–5 years and continue growing to 13% by year seven, confirming dynamic adjustment frictions (Section 5.2). Third, the coastal-inland divergence persists across all examined dimensions—industries, ownership types, products, and destinations—confirming that geographic position, not firm characteristics, determines infrastructure benefits (Section 5.3). Section 5.4 addresses endogeneity concerns using instrumental variables.

5.1 Average Effects on Exports

We begin by testing Prediction 1: expressway expansion increases firm-level exports on average by reducing domestic transportation costs (τ_i^{road}), with effects declining as distance to the nearest

expressway increases. Critically, these average effects mask sharp regional divergence.

We estimate equations 17 and 18 to test this prediction. Equation (17) defines six treatment groups based on distance to the expressway network (0–10km, 10–20km, 20–30km, 30–40km, 40–50km, 50–60km), allowing us to trace out the distance gradient in treatment effects. Equation (18) estimates average effects using a binary treatment indicator for firms within 40km of new expressways. Table 2 presents distance-gradient estimates. Table 3 presents average treatment effects. Both specifications control for firm-product and country-year fixed effects, with standard errors clustered at the firm level.

Distance-Gradients. Table 2 presents estimates from equation (17). Column (1) uses the full sample. Columns (2)–(4) split the sample by region (Eastern, Central, and Western China), allowing treatment effects to vary with baseline port access. Figure 3 plots the distance-gradient coefficients from Column (1).

Results confirm Prediction 1’s expectation of steep distance gradients. Column (1) shows that expressway access increases exports significantly for firms within 40km of new connections, with effects strongest for firms 0–10km away. Effects decline monotonically with distance and become statistically indistinguishable from zero beyond 40km. Figure 3 illustrates this pattern visually. The steep gradient is consistent with the model’s prediction that τ_i^{road} declines sharply with distance, generating proportionally smaller export gains for more distant firms.

Table B3 in Appendix B confirms these results are robust to alternative fixed effects specifications. We progressively augment the baseline specification to account for product-specific preference trends (adding product-year fixed effects), country-specific product demand shifts (adding product-country-year fixed effects), and time-invariant firm-product-destination relationships (replacing firm-product with firm-product-country fixed effects). The estimates remain stable across all specifications, including the most demanding one that relies solely on variation across firms exporting the same product to the same destination in the same year.

Columns (2)–(4) reveal striking regional heterogeneity that confirms Prediction 1. Eastern China—home to major seaports (Shanghai, Ningbo, Shenzhen, Guangzhou)—experiences positive effects of 8–12% across all distance bands within 40km, with the largest gains (approximately 12%) for firms 0–10km away. In sharp contrast, Central and Western China experience negative ef-

fects of 4–7% and 10–15%, respectively, within their effective ranges. For Central China, negative effects are statistically significant within 20km; for Western China, within 10km. The geographic range of effects also varies: Eastern coastal firms benefit at greater distances (up to 40km), while inland effects attenuate rapidly, consistent with the model’s prediction that low- τ_i^{port} firms translate road improvements into larger delivered price reductions.

Importantly, the negative near-field effects in Central and Western China rule out measurement error explanations. Even inland firms within 10km of new expressways, those experiencing the largest τ_i^{road} reductions, see export declines. This pattern confirms the competitive externality mechanism: coastal firms’ cost reductions and market share expansions (equation (13)) reduce residual demand faced by inland exporters regardless of whether inland firms themselves gain expressway access.

Figure A5 in Appendix A provides granular geographic evidence. Positive treatment effects concentrate in coastal provinces with major international ports (Jiangsu, Zhejiang, Guangdong, Shanghai, Tianjin). Negative effects dominate in landlocked provinces (Henan, Shanxi, Anhui, Sichuan, Shaanxi). The spatial pattern reinforces that port accessibility (τ_i^{port}) determines whether firms translate road improvements into export gains.¹¹

Average Treatment Effects. Table 3 presents average treatment effects from equation (18). Panel A defines the treatment group as firms within 40km of new expressways, comparing them to firms 40–100km away. Panel B defines treatment at the county level (whether the county intersects with the expressway network). Panel C estimates the distance elasticity by regressing log exports on log distance to the nearest expressway. Column (1) reports the overall average effect. Columns (2)–(4) report effects by region.

Panel A, Column (1) shows that expressway access increases exports by 5.9% on average ($e^{0.057} - 1 = 0.059$).¹² However, this average masks substantial regional divergence. Columns (2)–(4) reveal that Eastern China experiences 6.5% export gains, while Central and Western China

¹¹ Within Eastern China, we examine heterogeneity by distance to coast. Interestingly, firms further inland within Eastern China benefit more from expressway access, likely because their port frictions are similar but the domestic transportation costs are higher, making marginal gains from improved connectivity more substantial. This pattern aligns with the model’s prediction that infrastructure improvements yield the greatest gains for firms facing the highest initial trade frictions. Full results available upon request.

¹² We convert semi-logarithmic coefficients to percentage effects using the transformation $e^\beta - 1$.

experience 6.8% and 15.0% declines, respectively. The 21.5 percentage point difference between Eastern gains and Western losses confirms the divergence effect predicted in Prediction 1.

These regional patterns align precisely with the model’s prediction. Equation (4) shows that reductions in τ_i^{road} translate into delivered price reductions proportional to $(1 - \tau_r^{\text{port}})$. Coastal firms with low τ_r^{port} achieve larger proportional price reductions, gaining competitive advantages. Equation (13) shows that as coastal firms expand market shares, inland firms lose residual demand even when they themselves gain infrastructure access—precisely the pattern we observe in Panel A, Columns (2)–(4).

Panel B shows that county-level treatment definition yields a smaller average effect (3.2% in Column 1), introducing downward bias compared to firm-level treatment (5.9% in Panel A). This attenuation arises because county-level treatment creates two forms of misclassification: some “treated” firms remain far from expressways within connected counties (diluting treatment intensity), while some “control” firms lie near county borders with expressway access (contaminating the control group).

Panel C estimates the distance elasticity. Column (1) shows that a 10% reduction in distance to the nearest expressway raises exports by 2.9% on average. Regional elasticities in Columns (2)–(4) reveal striking heterogeneity. Eastern China exhibits the expected negative elasticity (-0.033): Closer proximity to expressways increases exports. In sharp contrast, Central and Western China exhibit positive elasticities, with Western China showing a significant coefficient (0.059), indicating that firms closer to expressways experience export declines. This counter-intuitive pattern reflects the competitive externality mechanism: expressway expansion intensifies coastal firms’ competitive advantages, and inland firms located closer to new expressways may face stronger displacement effects from expanding coastal competitors compared to more remote inland firms that remain partially insulated from coastal competition.

Together, the distance gradient and average treatment effect results provide strong support for Prediction 1. Expressway expansion increases exports on average with steep distance gradients, but average effects mask dramatic regional heterogeneity driven by port access and competitive externality. The 21.5 percentage point divergence between Eastern gains and Western losses implies substantial spatial reallocation of export activity toward coastal regions.

5.2 Dynamic Effects

This section tests Prediction 2, which posits that export responses materialize gradually over a multi-year horizon due to adjustment frictions. We implement event-study designs that trace out treatment effects year-by-year relative to expressway connection timing. Figure 4 presents the baseline event study. We validate these results using two recent econometric methods robust to potential bias in staggered difference-in-differences designs: the De Chaisemartin and d’Haultfoeuille (2020) estimator and the Rambachan and Roth (2023) “Honest DID” approach. Figure A6 presents event studies separately by region, testing whether dynamic patterns vary with baseline port access as Prediction 1 suggests.

Event-Study Evidence. Panel (a) of Figure 4 plots event-study coefficients. Red dots report the estimates from equation (19); blue dots report De Chaisemartin and d’Haultfoeuille (2020) estimates. We normalize coefficients at $k = -1$ to zero and cluster standard errors at the firm level.

Three key patterns emerge. First, all pre-treatment coefficients are statistically indistinguishable from zero at the 95% confidence level for both estimators. Pre-treatment point estimates fluctuate narrowly around zero with no systematic drift, validating the parallel trends assumption underlying our difference-in-differences design. This pattern rules out differential pre-trends between treated and control firms.

Second, treatment effects emerge gradually. Effects remain close to zero in the first two post-treatment years ($k = 0, 1$), then become positive and statistically significant starting at $k = 3$. The delayed response is consistent with Prediction 2, which suggests that expressway expansion leads to a gradual accumulation of effects rather than an immediate discrete shift at the time of connection.

Third, effects continue to grow monotonically throughout the post-treatment period, reaching approximately 13% by $k = 7$. This sustained growth reflects three reinforcing mechanisms: (i) search costs and relationship-specific investments in forming connections with foreign buyers (Eaton et al., 2025); (ii) convex costs of expanding export capacity, and (iii) gradual learning about foreign market opportunities (Albornoz et al., 2012). These mechanisms generate cumulative effects rather than one-time level shifts, explaining the sustained upward trajectory observed in Panel (a).

The blue dots in Panel (a) show that [De Chaisemartin and d’Haultfoeuille \(2020\)](#) estimates closely track the Two-Way Fixed Effects (TWFE) estimates throughout the pre- and post-treatment periods. Both estimators yield flat pre-treatment coefficients and monotonically increasing post-treatment effects. The similarity confirms that treatment effect heterogeneity across cohorts does not bias our baseline findings. The [De Chaisemartin and d’Haultfoeuille \(2020\)](#) estimator accounts for potential negative weighting problems in staggered DID designs by isolating and aggregating cohort-specific effects, yet produces nearly identical point estimates to TWFE, validating our main specification.

While the flat pre-treatment coefficients support parallel trends, one might still worry that the upward post-treatment trajectory reflects small, undetectable violations of this assumption. Panel (b) of Figure 4 implements the [Rambachan and Roth \(2023\)](#) “Honest DID” approach, testing robustness to bounded deviations from parallel trends. The analysis shows treatment effects remain positive and significant even when allowing for modest violations of the parallel trends assumption, providing additional confidence in our baseline estimates.

Figure A6 also presents event studies separately by region. Panel (a) shows Eastern China exhibits sustained growth, while Panels (b) and (c) show that Central and Western China experience stagnant or slightly negative effects throughout the post-treatment period. The divergent dynamics provide additional evidence against differential pre-trends. If the upward trajectory simply reflected spurious trends, we would expect similar patterns across regions. Instead, we observe sustained growth in Eastern China versus stagnation in Central/Western China—precisely the pattern predicted by the model’s competitive externality mechanism. As coastal firms expand and capture market shares, competitive pressures intensify for inland firms, consistent with Prediction 1.

5.3 Additional Dimensions of Heterogeneity

Sections 5.1 and 5.2 documented that expressway effects vary sharply by geographic region, with coastal firms experiencing export gains while inland firms experience losses. This subsection examines whether the divergence patterns persist across other dimensions of firm heterogeneity. We investigate heterogeneity by industry characteristics, ownership structure, product types, and export destinations. These analyses assess the robustness of the core regional findings and provide

additional evidence that infrastructure benefits accrue disproportionately to firms best positioned to exploit international market access.

Table 4 presents estimates from equation (18) with the sample split along various dimensions. Panel A examines industry and ownership heterogeneity. Panel B examines product and destination heterogeneity. All specifications control for firm-product and country-year fixed effects, with standard errors clustered at the firm level. Table B4 in Appendix B reports triple-difference estimates that formally test whether treatment effects differ across subgroups.

5.3.1 Industry Characteristics and Ownership Structure

Manufacturing vs. Non-Manufacturing. Columns (1)–(2) in Panel A show that manufacturing firms, which constitute 99% of observations, experience 5.8% export increases. Non-manufacturing firms show larger point estimates (11.9%), but the triple-difference specification (Table B4 Column 1) indicates this difference is not statistically significant. The large manufacturing effect aligns with prior evidence that manufactured goods dominate trade flows and benefit disproportionately from transport cost reductions (Ghani et al., 2016; Hummels, 2007). Figure A7 shows that the largest impacts occur in textiles, non-metallic minerals, and equipment manufacturing.

Road Reliance. Industries vary in their reliance on road transportation. We classify industries as high or low road dependence based on transport intensity from the 2007 Chinese Input-Output tables.¹³

Columns (3)–(4) show significant positive effects for both high-reliance (9.3%) and low-reliance (3.0%) industries. Interestingly, the triple-difference estimate (Table B4 Column 2) indicates low-reliance industries benefit 9.5 percentage points more than high-reliance industries. This counter-intuitive result likely reflects that low-reliance industries (e.g., electronics) concentrate in Eastern coastal areas with superior port access, while high-reliance industries (e.g., textiles, mining) are more geographically dispersed.

¹³Following Wu et al. (2023), high road-reliant industries include mining, textiles, wood products, and transportation services—sectors with heavy logistics requirements for raw materials and finished goods. Low road-reliant industries include chemicals, pharmaceuticals, machinery, and electronics—sectors with higher value-to-weight ratios or more capital-intensive production.

Ownership Structure. Columns (5)–(7) in Panel A examine heterogeneity by ownership type. SOEs, despite representing only 4.2% of exporters, experience the largest gains (17.7%). Private firms, the dominant exporter group, experience 4.7% gains. Foreign firms show no significant average response.

These patterns reflect two forces. First, SOEs benefit from preferential access to subsidies and policy support (Brandt et al., 2012), amplifying their ability to exploit infrastructure improvements. Second, private firms demonstrate greater market responsiveness and technology adoption (Wang, 2023), enabling them to capitalize on new opportunities despite fewer policy advantages. Foreign firms, already possessing established export networks and sophisticated logistics, gain less marginal benefit from domestic road improvements. The triple-difference estimate (Table B4 Column 3) indicates private firms benefit 13.1 percentage points more than the combined SOE and foreign firm group.

5.3.2 Destinations, Product Types, and Trade Regimes

Export Destinations. Columns (1)–(2) in Panel B examine heterogeneity by destination. Exports to OECD countries increase 6.4%; exports to non-OECD countries increase 4.0%. The triple-difference estimate (Table B4 Column 4) indicates no significant difference between destination types. Continental breakdowns (Table B8) show positive effects for Asia, Europe, and North America, but negative effects for South America, Africa, and Oceania—regions with smaller trade volumes and less established shipping routes from China.

The lack of differential effects across OECD/non-OECD destinations suggests that expressway improvements reduce domestic transportation costs uniformly. This pattern is consistent with infrastructure improvements lowering the fixed costs of market entry, enabling firms to serve multiple destinations more efficiently (Manova and Zhang, 2012).

Product Type Effects. We classify products into capital goods, intermediate goods, and final goods using the BEC Revision 4 classification. Columns (3)–(5) in Panel B show that final goods exports increase significantly (7.4%), while capital and intermediate goods show positive but smaller and statistically insignificant effects. Although the triple-difference estimate (Table B4 Column 5) indicates the difference is not statistically significant at conventional levels, the point

estimates suggest final goods benefit more from reduced transport costs, consistent with [Hummels and Skiba \(2004\)](#), who show that high-value products are particularly sensitive to shipping cost changes.

Processing and Ordinary Exports. We examine whether expressway expansion affects processing trade (assembly using imported inputs) differently from ordinary trade. Using data through 2006 (when trade regime classifications are reliable), Columns (6)–(7) show that processing exports increase significantly (20.0%), while ordinary exports show no significant response. The triple-difference estimate (Table [B4](#) Column 6) confirms that processing exports benefit 19.4 percentage points more than ordinary exports.

This pattern reflects that processing trade firms, concentrated in coastal export processing zones, benefit doubly from expressway expansion: improved access to ports for importing intermediate inputs and exporting finished goods. Table [B12](#) further shows that dual-regime firms (engaging in both processing and ordinary trade) drive the positive effects, suggesting expressway expansion enables operational diversification for sophisticated exporters.

5.3.3 Regional and Ownership Decomposition

Tables [B5](#) through [B11](#) in Appendix B present detailed decompositions by region and ownership type for each heterogeneity dimension. A remarkably consistent pattern emerges: positive effects concentrate in Eastern China while Central and Western regions experience negative effects across all dimensions examined. For example, manufacturing firms in Eastern China experience 6.1% export gains while Western manufacturing firms experience 13.2% declines (Table [B5](#)). Processing trade shows particularly stark patterns: 24.7% in Eastern China versus negative 18.9% in Western China (Table [B11](#)). Critically, negative inland effects persist even for firms with structural advantages. SOEs experience 15.1% export declines in Western China despite policy advantages (Table [B7](#)), while foreign firms suffer 9.4% losses despite established global networks.

5.3.4 Synthesis

The heterogeneity analysis confirms two key findings. First, considering other dimensions of firm heterogeneity, the largest gains accrue to firms best positioned to exploit global market in-

tegration: SOEs with policy support (17.7%), private firms with “advantages of backwardness” (4.7%), processing trade firms (20.0%), and final goods exporters (7.4%). The processing trade gain, largest across all categories, reflects these firms’ dual dependence on port access for both imported inputs and exported outputs.

Second, the regional pattern persists across every dimension examined. This consistency demonstrates that geographic position relative to international markets, not firm characteristics, primarily determines infrastructure benefits. This universality of negative inland effects provides the strongest evidence for Prediction 1: as coastal firms with low τ_r^{port} exploit infrastructure improvements to reduce delivered prices and expand market shares, inland firms with high τ_r^{port} lose residual demand even when gaining their own expressway access.

5.4 Instrumental Variable Estimation

Our baseline identification assumes expressway placement is uncorrelated with unobserved export growth determinants. This could be violated if the government prioritized expressway construction in regions with strong export potential, preferentially connecting export-intensive cities, or if reverse causality exists whereby export growth itself attracted infrastructure investment. Omitted variable bias could also arise if political or economic factors that influence both expressway placement and firm performance remain uncontrolled.

We implement two instrumental variables strategies to address these concerns. First, following Faber (2014), we construct a least-cost path spanning tree network that connects target cities (population exceeding 200,000) while minimizing construction costs based on terrain slope and land cover.¹⁴ This counterfactual network is determined by geographic and engineering constraints rather than contemporaneous economic conditions. Second, we instrument using the Ming Dynasty (1368–1644) Courier Road network from 1587, exploiting the historical persistence of transportation corridors while ensuring exogeneity to modern economic shocks. Figure 5 compares the actual 2013 expressway network with these two instrumental networks, showing substantial alignment in both cases, though the modern network is considerably denser.

Since these instruments do not capture the temporal rollout of expressway construction, we

¹⁴Target cities are identified from the Chinese Expressway Network Construction Plan (2004). Construction costs are calculated using 30m × 30m grid cell terrain data. Figure A7 displays the construction cost distribution map.

define treatment at the county level: a county is treated if connected to the instrumental network. To introduce temporal variation, we follow [Frye \(2024\)](#) and [Wu et al. \(2023\)](#) by modeling annual construction progress using each city’s yearly road construction budget. Specifically, the treatment variable equals the interaction between county connectivity to the instrumental network and the city’s cumulative budget share in each year.

Table 5 presents instrumental variables estimates. Panel A uses the least-cost spanning tree instrument for the full sample. Panel B uses the same instrument but restricts to non-target cities, addressing concerns that target city selection itself may be endogenous. Panel C uses the historical courier road instrument, excluding firms in provinces not covered by the historical network (Inner Mongolia, Jilin, Heilongjiang, Tibet, Qinghai, and Xinjiang), following [Banerjee et al. \(2020\)](#). Column (1) of each panel reports full-sample estimates; Columns (2)–(4) report results separately for Eastern, Central, and Western China.

The regional patterns align closely with baseline findings: across all three panels, expressway connectivity generates significant positive effects in Eastern China but negative or insignificant effects in Central and Western China. First-stage Kleibergen-Paap rk Wald F-statistics exceed conventional weak instrument thresholds in all specifications, rejecting weak instrument concerns.

The IV estimates are substantially larger than the baseline estimates in Table 3—approximately 10 times larger, implying treatment effects of 40–60% for complier counties. While these magnitudes may seem large, they are consistent with prior Chinese expressway studies ([Faber, 2014](#); [Wu et al., 2023](#)) and reflect local average treatment effects for complier locations experiencing the most dramatic connectivity improvements. First, the IV identifies local average treatment effects for complier counties—those connected because they lie on the least-cost or historical routes. These compliers likely experience more dramatic connectivity improvements than the average treated county, generating larger treatment effects. Second, if expressways were endogenously placed to address regional inequality or political objectives, baseline estimates suffer from attenuation bias. The IV corrects this by isolating variation driven by geographic constraints. Third, network density matters: the actual expressway network is far denser than either instrumental network, particularly the sparse Ming Dynasty roads. The IV, therefore, concentrates identification on locations where connectivity changes are most dramatic, explaining why Panel C yields somewhat larger coefficients than Panel A.

Critically, despite magnitude differences, the regional pattern remains consistent across all specifications: positive effects in Eastern China, negative effects in Central and Western China. This consistency confirms that the competitive externality mechanism reflects causal infrastructure impacts rather than endogenous placement artifacts. Even under the cost-minimizing design (Panel A) or historical routing (Panel C), infrastructure concentrates benefits in coastal regions while generating inland losses. The larger IV magnitudes reinforce rather than contradict this finding: when identifying effects for complier counties experiencing dramatic connectivity changes, the same regional divergence emerges.

5.5 Robustness Checks

To verify the robustness of the baseline findings in Table 3, we conduct several sensitivity analyses addressing potential confounding factors and sample composition concerns. Table 6 presents results from four robustness checks.

Alternative Transportation Infrastructure. Columns (1)–(3) address concerns that other transportation infrastructure—railways, waterways, and urban road networks—may confound the estimated expressway effects. Column (1) excludes firms in railway hub cities, isolating expressway effects from rail network impacts. Column (2) excludes firms in inland port cities, controlling for waterway access.¹⁵ Column (3) excludes firms in urban districts (*shi xia qu*), where dense local road networks may facilitate expressway access independently. The estimated coefficients remain positive and statistically significant across all specifications, with magnitudes nearly identical to the baseline estimate in Table 3 Column (1). This consistency confirms that expressway expansion—not other transportation modes—drives the observed export effects.

Intermediary Firms. Column (4) examines export effects for intermediary firms—trading companies that purchase and resell goods without direct production involvement. These firms were excluded from the baseline analysis because measuring their distance to expressways is ambiguous: the relevant distance may be to suppliers’ production facilities rather than to intermediaries’ registered addresses. The results show a negative but statistically insignificant coefficient, suggest-

¹⁵Railway and waterway hub cities are identified from the 2004 National Expressway Network Plan.

ing expressway expansion does not significantly affect intermediary exports.¹⁶ This null finding is consistent with intermediaries relying primarily on established logistics networks and trading relationships rather than direct access to new transportation infrastructure.

Balanced Panel. Column (5) restricts the sample to incumbent firms—those continuously active from 2001 through 2013—testing whether the baseline results hold for a stable set of exporters.¹⁷ The estimated coefficient (0.075) is similar in magnitude and significance to the baseline estimate in Table 3, confirming that the positive export effects documented in Section 5.1 are not driven solely by sample composition changes.

6 Mechanisms

Section 5 documented that expressway expansion generates a 21 percentage point coastal-inland difference, with Eastern China gaining 6.5% while Western China loses 15.0%. This section investigates the mechanisms driving these divergent outcomes, testing Predictions 3 and 4. We find that coastal firms pass through cost savings as 5.4% FOB price reductions, while inland firms raise prices by 7.7% due to competitive pressures and factor outflows. Market reallocation operates through three complementary channels: Eastern firm entry declines as incumbents consolidate (consistent with within-firm expansion), county-level market concentration falls across all regions (reflecting local competition intensification), and resource allocation improves in Eastern China but deteriorates inland by 3.5–4.6%.

6.1 Export Price Pass-Through

Prediction 3 posits that expressway expansion reduces export prices by lowering domestic transportation costs (τ_i^{road}), with price declines serving as an intermediate mechanism linking infrastructure improvements to quantity increases (Prediction 1). The model further predicts that price effects should vary by region: coastal firms with low baseline port costs (τ_r^{port}) should experience larger price declines, whereas inland firms may face smaller price declines or even price increases

¹⁶Regional heterogeneity analysis shows similar null effects across Eastern, Central, and Western regions.

¹⁷We begin in 2001 rather than 2000 because the number of observations in 2000 is only 28% of that in 2001.

due to competitive pressures arising from coastal expansion.

We estimate equation (17) using log FOB export prices as the dependent variable. Table 7 presents distance-gradient estimates, with Column (1) reporting full-sample results and Columns (2)–(4) reporting regional results for Eastern, Central, and Western China. Table B13 presents average treatment effects and distance elasticities by replicating Table 18. All specifications include firm-product and country-year fixed effects, with standard errors clustered at the firm level.

Table 7 Column (1) shows that expressway access reduces export prices on average, with effects strongest within 20km of new connections. Columns (2)–(4) reveal striking regional heterogeneity, directly confirming Prediction 3 nuanced predictions.

Column (2) shows that Eastern China firms experience FOB price reductions, concentrated among firms within 20km of expressways. Table B13 Column (2) quantifies the average effect at 5.4%. This pattern is consistent with τ_i^{road} pass-through: as domestic transportation costs decline, coastal firms with low τ_r^{port} translate these savings into competitive delivered prices for international buyers.

Columns (3)–(4) reveal an apparent contradiction: Central and Western China firms experience FOB price increases. Table B13 Columns (3)–(4) show average increases of 7.7% and 4.3%, respectively. At first glance, this seems inconsistent with cost reduction. However, two complementary mechanisms explain this pattern.

First, the competitive externality mechanism (equation (13)) operates across regions. When coastal firms reduce delivered prices and expand export market shares (equation (13)), inland firms face declining residual demand in international markets—a competitive externality well-documented in trade literature (Melitz and Ottaviano, 2008). To maintain profitability on their shrinking export base, inland firms may raise export prices—a rational response to demand contraction that further reduces their competitiveness. This creates a vicious cycle: higher prices lead to lower quantities, which necessitate further price increases to cover fixed costs.

Second, local market structure changes raise production costs within inland regions. Simultaneously, expressway-induced clustering near new connections intensifies competition for local inputs—labor, land, and intermediates—within inland regions. Unlike coastal areas with thick factor markets and established supply chains (Combes et al., 2008; Gaubert, 2018), inland regions have thin input markets where increased demand generates sharp price increases. Additionally,

factors flow outward toward higher-return coastal opportunities (Moretti, 2010), further tightening inland input availability. These rising production costs force inland firms to raise export prices even as road access improves.

The distance gradient in Table 7 confirms these mechanisms operate locally: price increases concentrate among inland firms within 20km of expressways, where clustering effects and factor competition are strongest. For inland firms, the combination of competitive pressure from coastal expansion and rising local input costs dominates any direct transportation cost savings from expressway access. These patterns remain robust when defining treatment at the county level (Table B13 Panel B) or estimating distance elasticities (Panel C).

The regional pattern in price responses mirrors the regional pattern in quantity responses (Section 5.1), confirming Prediction 3 that FOB price changes serve as intermediate outcomes linking infrastructure improvements to export quantities. Coastal firms convert cost savings into lower prices, expanding demand through standard price elasticity. Inland firms face rising costs and competitive pressure, contracting exports even when gaining expressway access. This price heterogeneity provides direct evidence for the competitive externality mechanism. If expressway expansion simply reduced transportation costs uniformly across space, all firms would experience price declines. Instead, the divergent price responses—declines in coastal regions, increases inland—confirm that competitive reallocation dominates mechanical cost reduction for inland producers.

6.2 Market Reallocation

Having documented asymmetric price responses, we now investigate the market structure changes underlying these patterns. Prediction 4 posits that expressway expansion reallocates export market shares toward coastal firms through entry-exit dynamics, concentration changes, and resource allocation shifts. We examine these three complementary channels.

6.2.1 Firm Entry and Exit

We first examine extensive margin adjustments at multiple levels: firm entry and exit, product diversification, and destination market expansion. We construct a firm-year dataset and define

binary indicators for entry (firm begins exporting in year t) and exit (firm stops exporting in year t). We also count the number of eight-digit HS products and destination countries for each firm annually. Table 8 presents estimates from equation (18). All specifications include firm and year fixed effects, with standard errors clustered at the firm level.

Entry and Exit. Columns (1)–(4) show that expressway expansion reduces entry by 3.6% overall, with effects concentrated in Eastern China. Central China experiences modest entry increases. Columns (5)–(8) show slight exit rate increases, particularly in Central and Western China.

Eastern China experiences 4.5% lower entry, suggesting incumbents consolidate market positions. This reflects not market saturation but rising productivity thresholds: as established exporters leverage improved infrastructure and existing buyer relationships, entrants face higher competitive barriers.

In Central China, entry rises 2.1% while exit also increases, indicating market churning as competitive pressures intensify. The simultaneous increase in entry and exit reflects Schumpeterian creative destruction (Schumpeter, 1942): some firms attempt to exploit new connectivity while others fail under intensified competition from coastal producers.

Product and Destination Diversification. Beyond firm entry and exit, expressway expansion affects product portfolios and destination markets. Columns (9)–(12) examine the number of products exported, while Columns (13)–(16) examine the number of destination countries.

Columns (9)–(12) show that expressway expansion increases the number of products by 4.4% overall, with effects concentrated in Eastern China (8.2%). Columns (11)–(12) show Central and Western China experience significant declines in product counts (-6.2% and -12.5%, respectively).

Columns (13)–(16) show similar patterns for destination markets: overall increases of 6.6%, concentrated in Eastern China (+10.1%). Columns (15)–(16) show Central and Western China experience destination count declines (-3% and -10.7%, respectively).

These patterns confirm that expressway expansion generates divergent effects across regions. In Eastern China, despite reduced entry rates, incumbent firms expand their product portfolios and destination markets, reflecting sequential learning and market experimentation enabled by improved connectivity (Albornoz et al., 2012). The combination of lower entry and broader product-

destination scope suggests established Eastern exporters consolidate market positions while deepening their international engagement. In contrast, Central and Western China experience market churning: both entry and exit rates increase as firms experiment with new export opportunities, but face intensified competition. However, this churning occurs alongside portfolio contraction, with declining product counts and destination counts. This pattern suggests competitive pressures from coastal expansion force inland firms to concentrate on core products and markets rather than expanding their export scope, even as some firms attempt entry while others exit.

Table B14 also decomposes these patterns by ownership. Private firms—the dominant exporter group—drive the positive effects, experiencing entry increases, exit declines, and expansion in both products and destinations. SOEs show increased both entry and exit, reflecting policy-driven experimentation combined with market discipline, but limited product or destination diversification. Foreign firms experience declining entry and rising exit, with mixed effects on product and destination counts, suggesting multinational enterprises find China’s export platform less attractive as domestic firms consolidate competitive advantages.

6.2.2 Market Concentration

Having established that expressway expansion affects firm entry and exit differently across regions, we examine how these extensive-margin adjustments translate into market structure changes. We measure market concentration using the Herfindahl-Hirschman Index (HHI) at county-year and county-industry-year levels, where a lower HHI indicates more firms competing in the market and thus intensified competition. We estimate equation (18) using county-year (county-industry-year) level data with city and year (industry-year) fixed effects. Table 9 Panel A presents results.

Column (1) shows that expressway expansion reduces HHI by 6.8% at the county-year level, indicating increased firm participation and intensified competition in connected counties. Columns (2)–(4) reveal that effects vary by region: Eastern China experiences a 5% decline, Central China a 7.2% decline, and Western China the largest decline at 8.8%. At the county-industry-year level (Columns 5–8), the pattern is similar: overall HHI declines by 6.5%, with Eastern China by 7%, Central China by 5.3%, and Western China by 5.2%.

It is important to note that the HHI results use county-level treatment definitions (whether a county is connected to the expressway network), while the entry-exit analysis in Table 8 uses

firm-level treatment based on distance within 40km. Under the county-level treatment definition, Eastern China experiences both entry and exit declines (entry coefficient: -0.014; exit coefficient: -0.013), suggesting relative stability in the number of active exporters rather than the entry reduction documented under distance-based treatment.

The declining HHI in Eastern China, despite stable or slightly declining firm counts, reflects changes in market share distribution rather than increases in firm numbers. When HHI declines without substantial entry, it indicates that existing market shares are becoming more evenly distributed—smaller firms are expanding relative to larger firms, or dominant firms are losing market share to competitors. This pattern is consistent with expressway expansion enabling smaller or peripheral firms within connected counties to compete more effectively, reducing the market dominance of previously-leading firms even as the total number of participants remains relatively stable.

The regional pattern provides additional insights. Western and Central regions experience the largest HHI declines (8.8% and 7.2% at the county-year level) alongside entry-exit churning under distance-based treatment, suggesting substantial market restructuring in these regions. The large HHI declines reflect both increased firm participation in previously-underserved areas and redistribution of market shares among existing participants. However, this local market expansion and increased competition occur alongside aggregate export quantity declines (Section 5.1), confirming that inland counties experience intensified local competition while losing export market shares to coastal competitors.

The disconnect between local market structure (declining HHI indicating more competition) and export outcomes (declining quantities) underscores the competitive externality mechanism. Expressway expansion intensifies competition within inland counties as more local firms participate in export markets, reducing local concentration. However, this increased local participation fails to translate into export success because coastal firms simultaneously expand and capture market shares in international destinations. Inland firms face a dual challenge: intensified local competition for resources and customers, combined with intensified external competition from more productive coastal exporters.

6.2.3 Resource Misallocation

Finally, we assess the efficiency implications of expressway-induced reallocation by examining resource misallocation. Following [Hsieh and Klenow \(2009\)](#), we measure misallocation as the variance of log revenue productivity (TFPR) across firms within a county-year or county-industry-year unit.¹⁸ Higher variance indicates greater misallocation as resources fail to flow toward the highest-value uses. We estimate equation (18) using county-year (county-industry-year) level aggregated data, defining treatment as whether a county is directly connected to the expressway network. All specifications include city and year (or industry-year) fixed effects, with standard errors clustered at the county level. Table 9 Panel B presents results.

Column (1) shows that expressway expansion increases misallocation by 1.5% at the county-year level overall. However, Columns (2)–(4) reveal striking regional heterogeneity. Eastern China experiences a 2.4% decline in misallocation, indicating improved resource allocation as expressway expansion facilitates resource flows toward high-productivity exporters with strong port access. In sharp contrast, Central and Western China experience misallocation increases of 3.6% and 4.7%, respectively, reflecting deteriorating allocation as factors flow out toward coastal regions. At the county-industry-year level (Columns 5–8), the regional divergence in misallocation persists: Eastern China -1.2% (insignificant), Central China 2.6%, Western China 1.4% (insignificant).

This divergence in misallocation patterns provides direct evidence for the spatial reallocation mechanism predicted by the competitive externality framework (equation (13)). When expressway expansion reduces τ_i^{road} for firms across all regions, resources reallocate toward areas with lower τ_r^{port} —Eastern China—where returns to export-oriented production are highest. This reallocation improves allocative efficiency in receiving regions as resources flow to high-productivity exporters, but deteriorates efficiency in sending regions as factor outflows leave remaining firms operating with distorted factor combinations.

The improvement in resource allocation in Eastern China operates through several channels. First, expressway connections enable factors, particularly skilled labor and capital, to reallocate toward high-productivity exporters within the region, reducing the dispersion of marginal revenue products across firms. Second, as documented in Section 6.2.2, market shares redistribute toward more productive firms even as aggregate concentration declines, suggesting that resources flow to

¹⁸Appendix D provides details on TFPR estimation following [Hsieh and Klenow \(2009\)](#).

efficient users despite increased local competition. Third, both firm entry and exit decline slightly in Eastern China under county-level treatment, indicating a relatively stable firm composition that allows resources to allocate efficiently among established producers rather than being absorbed by entry-exit churning.

In contrast, the deterioration in resource allocation in Central and Western China reflects multiple distortions created by factor outflows and competitive pressures. First, resource outflows to coastal regions create factor imbalances: firms lose access to skilled labor and capital that migrate coastward, forcing remaining firms to operate with suboptimal and heterogeneous factor combinations. This generates increased dispersion in marginal revenue products across firms—the defining characteristic of misallocation in the Hsieh-Klenow framework. Second, increased local competition for remaining inputs generates factor price distortions, with firms facing heterogeneous input prices that prevent efficient allocation. Third, reduced export competitiveness documented in Section 5.1 creates output distortions as some firms contract while others expand to fill local niches, generating productivity dispersion.

6.3 Synthesis

The mechanism analysis provides strong support for Predictions 3 and 4. Export price pass-through operates as predicted: coastal firms experience price declines that facilitate quantity expansion, while inland firms experience price increases reflecting competitive pressures and factor outflows. Market reallocation operates through entry/exit dynamics, market concentration changes, and resource allocation shifts, all favoring coastal regions at the expense of inland areas.

Taken together with the quantity and heterogeneity results (Prediction 1) and dynamic adjustment evidence (Prediction 2), the findings confirm the conceptual framework’s central mechanism: expressway expansion generates competitive advantages for firms with superior port access, triggering spatial reallocation of export activity through price competition and market share redistribution. The consistency of this pattern across quantity effects, price pass-through, entry–exit dynamics, market concentration, and resource allocation provides converging evidence that geographic position relative to international markets determines infrastructure benefits through the competitive externality channel.

7 Conclusion

Transportation infrastructure represents one of the largest categories of public investment in developing countries, yet its effects on international trade and spatial inequality remain poorly understood. This paper exploits China’s massive expressway expansion—from 16,000km to 104,500km during 2000–2013—to examine how domestic infrastructure improvements affect firm-level export performance and regional divergence. Combining comprehensive customs data with precise geospatial measurements of annual network expansion, we identify causal effects using difference-in-differences and instrumental variables approaches.

We document three main findings. First, expressway expansion generates sharply divergent regional effects. Coastal firms experience 5.9% export increases, while inland firms experience declines of 6.8–15.0%, creating a 21 percentage point coastal-inland difference. This divergence persists across virtually all dimensions examined—industries, ownership types, products, and destinations—confirming that geographic position relative to international markets, not firm characteristics, determines infrastructure benefits.

Second, we develop a simple framework organizing these findings through a competitive externality mechanism. When expressway expansion reduces domestic transportation costs, firms differ in translating savings into competitiveness based on port access. Coastal firms with low port costs convert road improvements into larger cost reductions, enabling price declines that expand market shares. Inland firms with high port costs experience modest gains and lose residual demand as coastal competitors expand. The framework generates testable predictions about regional divergence, dynamic adjustment, price pass-through, and market reallocation—all confirmed empirically.

Third, we document mechanisms driving these outcomes. Export prices decline 5.4% in Eastern China but increase 7.7% in Central China, confirming asymmetric pass-through. Market reallocation operates through entry-exit dynamics (Eastern entry declines 4.5% as incumbents consolidate; inland regions experience churning), concentration changes (county-level HHI declines 5–9% across all regions despite aggregate inland export losses), and resource allocation shifts (allocation improves 2.4% in Eastern China but deteriorates 3.6–4.7% inland).

These findings carry important policy implications. Standard infrastructure evaluations focus-

ing on aggregate effects would conclude that expressway expansion substantially increased exports. However, this average masks profound regional divergence. The competitive externality mechanism demonstrates that domestic infrastructure improvements do not uniformly benefit all regions—geography determines whether road improvements translate into gains. These findings challenge conventional policy approaches. Policymakers cannot assume that infrastructure benefits are distributed uniformly across space. When complementary inputs (in our case, port access) distribute unevenly, infrastructure improvements that appear efficiency-enhancing on average can simultaneously exacerbate regional inequality through competitive reallocation. Effective spatial development policy may require coordinated investments in multiple complementary inputs or explicit redistribution mechanisms to ensure benefits reach disadvantaged regions, balancing aggregate efficiency gains against distributional consequences.

Several questions remain for future research. First, a comprehensive welfare analysis incorporating domestic markets and factor mobility could quantify net benefits beyond exports. Second, extending to services trade and domestic commerce would test whether coastal-inland divergence generalizes to other sectors. Third, examining longer-run channels—technology diffusion, agglomeration economies—could illuminate whether initial advantages compound over time. More broadly, our findings suggest that evaluating infrastructure based solely on aggregate effects may systematically underestimate distributional consequences. When complementary inputs distribute unevenly across space, infrastructure creates winners and losers through competitive externalities. Effective spatial policy requires explicitly accounting for geographic heterogeneity and balancing efficiency against equity.

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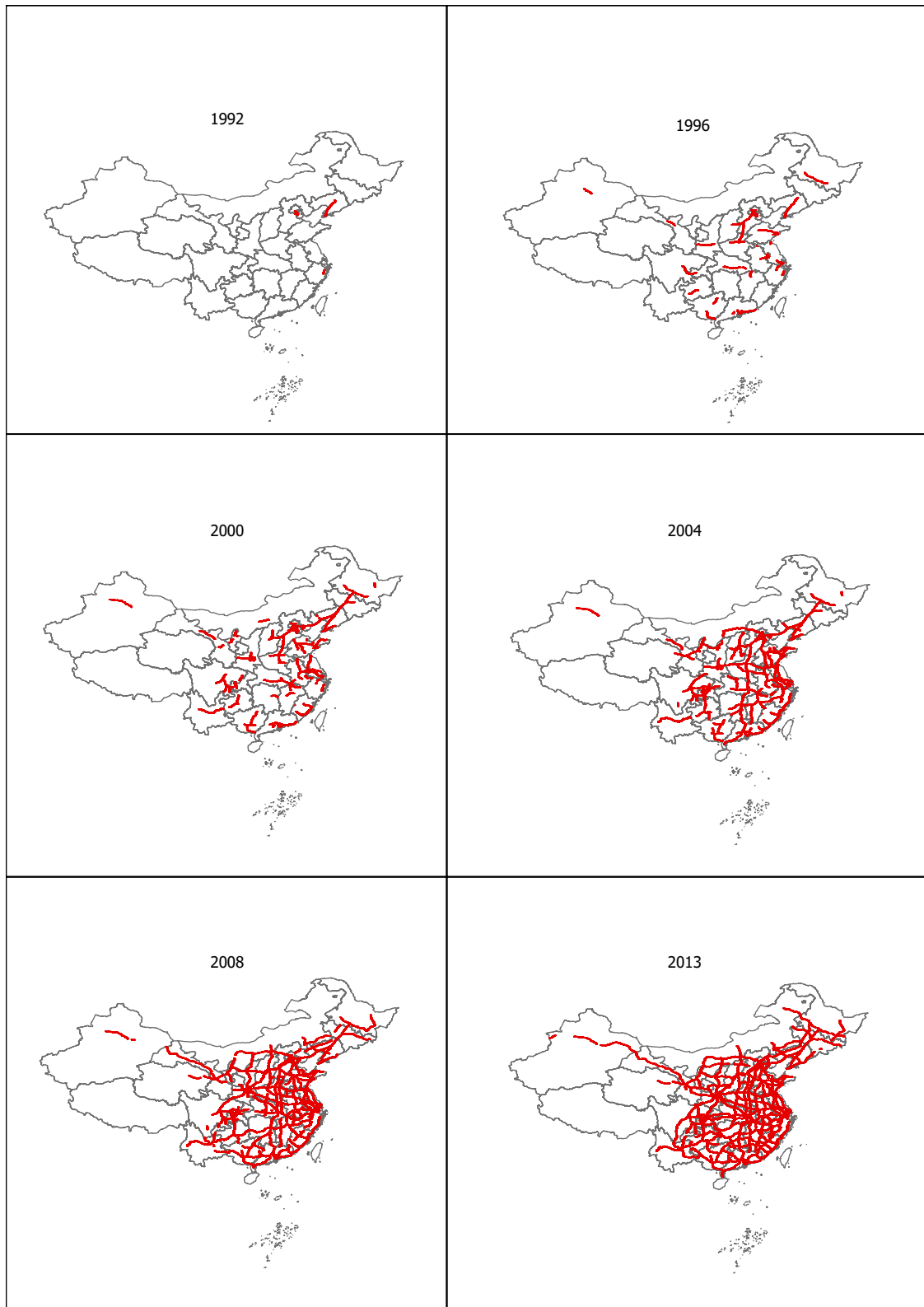
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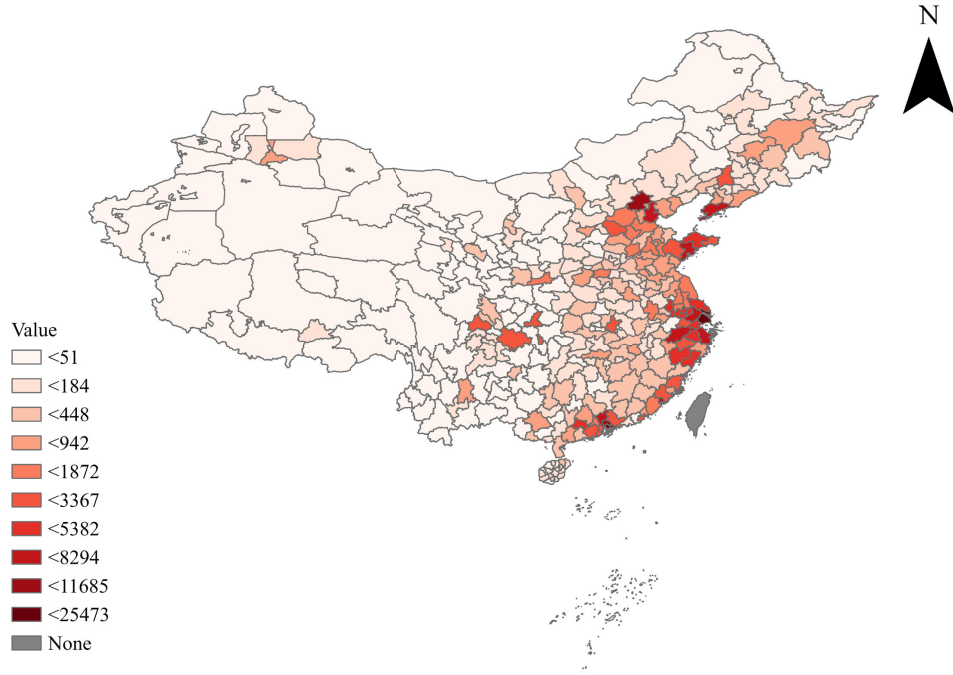
Figures

Figure 1: China's National Expressway Network, 1992–2013

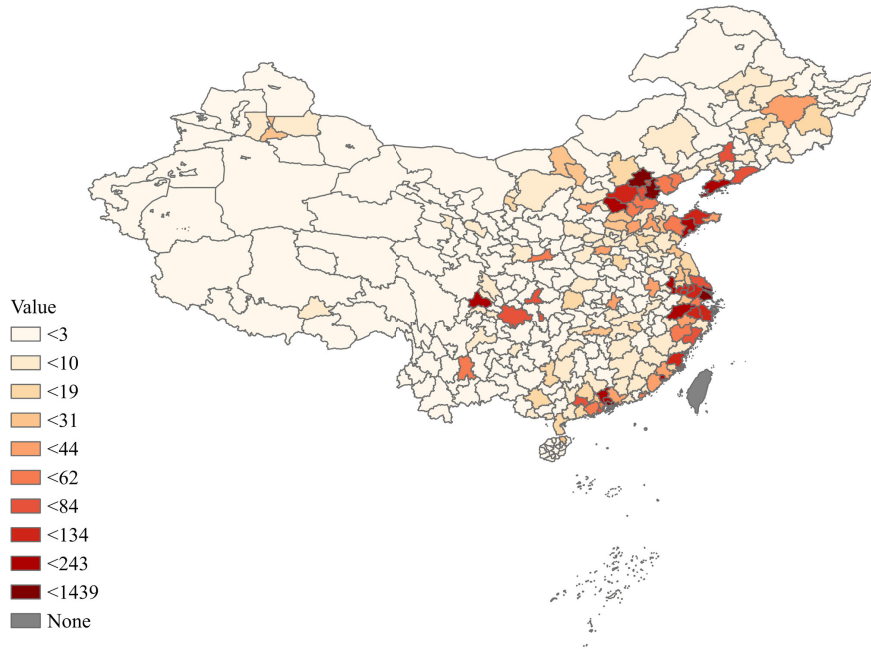


Notes: Each panel displays the national trunk expressway network (red lines) in the indicated year, overlaid on provincial boundaries (gray lines). Total network length grew from approximately 1,000km in 1992 to over 104,000km in 2013. Expressway shapefiles are manually digitized from annual transportation atlases published by SinoMaps Press and other major publishers (see Table B1).

Figure 2: Spatial Distribution of Exporters and Product Varieties, 2013



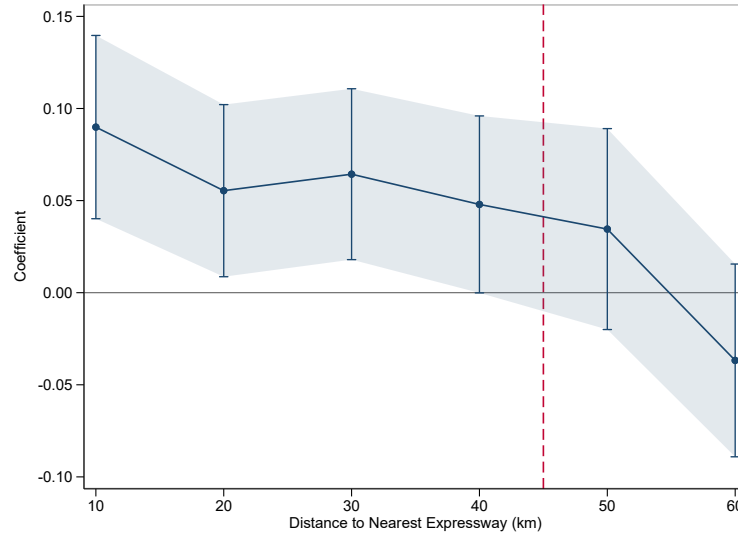
(a) Distribution of Firms



(b) Distribution of 8-digit Products

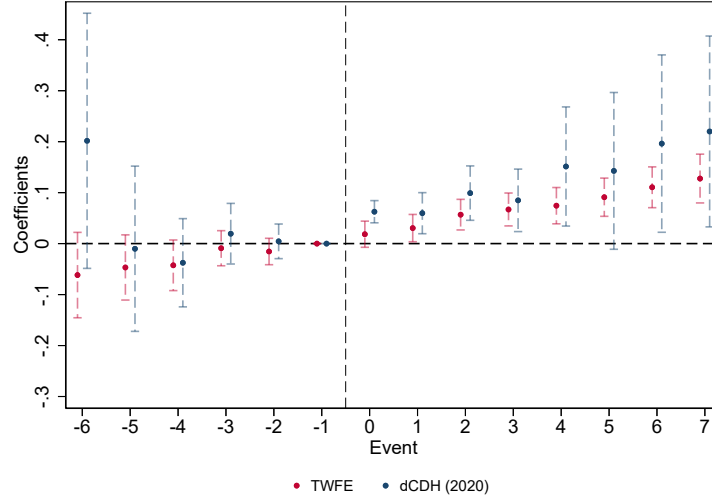
Notes: Panel (a) shows the number of exporting firms per city in 2013, and Panel (b) shows the number of distinct HS 8-digit products exported per city. Classification intervals are determined using the Natural Breaks (Jenks) method. Cities shaded in darker tones indicate higher firm counts or greater product diversity. Data are from the Chinese Customs Trade Statistics.

Figure 3: Effects of Expressway Proximity on Exports by Distance Band

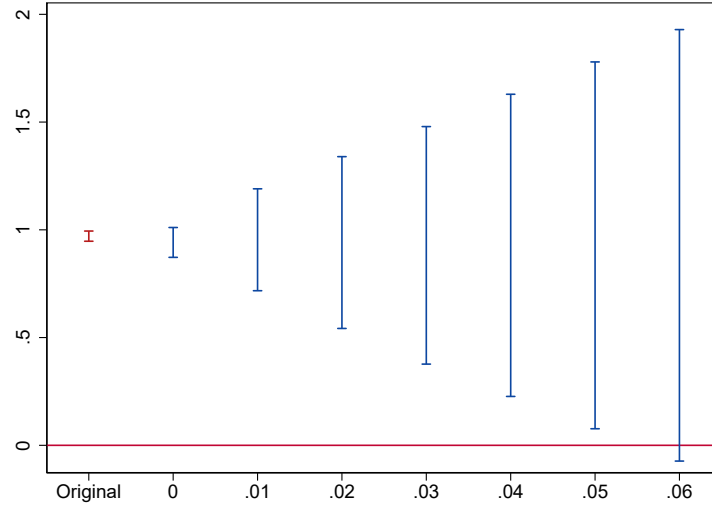


Notes: This figure plots estimated coefficients from equation (17), where the dependent variable is the logarithm of export amounts. Each point represents the effect of being located within a given 10km distance band from the nearest expressway, relative to firms located 60km or more away (the omitted reference group). The red dashed line indicates the 40km threshold used in the baseline specification (equation (18)). Shaded areas represent 95% confidence intervals. All specifications include firm–product and country–year fixed effects, with standard errors clustered at the firm level.

Figure 4: Event-Study Estimates of Expressway Proximity on Exports



(a) Event-Study Estimator



(b) Honest DID Estimator

Notes: Panel (a) reports event-study estimates of the effect of expressway access on firm-level exports, based on equation (19). The red dots show estimates from the standard two-way fixed effects (TWFE) estimator, and the blue dots show estimates using the method of [De Chaisemartin and d'Haultfoeuille \(2020\)](#), which accounts for heterogeneous treatment effects under staggered adoption. The omitted reference period is $k = -1$. Vertical bars denote 95% confidence intervals. Panel (b) presents a sensitivity analysis using the Honest DID approach of [Rambachan and Roth \(2023\)](#). The horizontal axis represents the maximum allowable deviation (Δ) from parallel trends per period, and the vertical axis reports the corresponding treatment effect estimates with 95% confidence intervals. The "Original" point denotes the baseline estimate from Panel (a).

Figure 5: Instrumental Variable Networks and the 2013 Expressway System



(a) Ming Dynasty Courier Road Network



(b) Least-cost Spanning Tree Network

Notes: Panel (a) overlays the Ming Dynasty (1587) courier road network (red) on the 2013 expressway network (blue). The courier road shapefile is sourced from [Berman and Zhang \(2017\)](#). Panel (b) overlays the least spanning tree network (red) on the 2013 expressway network (blue). The least-cost network is constructed using 30m-resolution elevation and land cover data to minimize construction costs while connecting all cities with populations exceeding 200,000.

Tables

Table 1: Summary Statistics for the Customs Sample

	Group	2000-2013			2000			2013		
		Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.
Distance to Nearest Expressway (km)	All	1,597,587	12.819	13.990	15,137	16.008	16.996	179,225	11.685	12.243
	Eastern	1,445,632	12.711	13.709	14,074	15.916	16.700	161,443	11.686	12.139
	Central	104,654	15.124	16.622	670	20.277	20.927	12,754	12.291	12.869
	Western	47,301	10.994	15.485	393	12.024	18.642	5,028	10.126	13.703
	SOEs	70,190	11.737	14.107	1,246	16.552	19.677	4,299	9.542	10.315
	FFs	799,509	12.688	14.081	13,029	15.416	16.121	63,798	11.259	12.294
	PFs	727,888	13.066	13.870	862	24.175	22.661	111,128	12.013	12.266
ln(amount)	All	35,786,072	7.377	3.289	471,223	5.422	3.893	4,121,678	6.840	3.495
	Eastern	32,836,500	7.381	3.274	455,684	5.466	3.889	3,665,682	6.848	3.497
	Central	1,689,358	7.460	3.652	10,607	4.224	3.811	242,200	6.795	3.792
	Western	1,260,214	7.153	3.142	4,932	3.893	3.715	213,796	6.753	3.087
	SOEs	4,894,046	7.494	3.323	20,474	4.531	3.945	335,737	6.560	3.652
	FFs	15,054,912	7.295	3.456	443,355	5.454	3.885	1,539,060	6.592	3.699
	PFs	15,837,114	7.418	3.109	7,394	5.958	3.892	2,246,881	7.051	3.309
ln(price)	All	35,786,072	6.454	2.384	471,223	2.750	3.056	4,121,678	7.232	2.390
	Eastern	32,836,500	6.441	2.358	455,684	2.704	3.043	455,684	7.229	2.361
	Central	1,689,358	6.628	2.769	10,607	3.965	3.040	10,607	7.349	2.795
	Western	1,260,214	6.547	2.495	4,932	4.340	3.266	4,932	7.149	2.373
	SOEs	4,894,046	6.358	2.543	20,474	3.977	3.342	20,474	7.467	2.796
	FFs	15,054,912	6.589	2.348	443,355	2.698	3.031	443,355	7.422	2.308
	PFs	15,837,114	6.355	2.362	7,394	2.458	2.957	7,394	7.067	2.367

Notes: This table reports summary statistics for the customs sample (Sample A), spanning 2000 to 2013, with separate snapshots for 2000 and 2013. The unit of observation is at the HS 8-digit product-firm-country-year level. “Distance to Nearest Expressway” measures the geodesic distance (in kilometers) from a firm’s geocoded address to the nearest expressway entrance. Subgroup statistics are reported by region (Eastern, Central, and Western China) and by ownership type (state-owned enterprises (SOEs), foreign firms (FFs), and private firms (PFs)). All monetary variables are deflated to the year 2000.

Table 2: Effects of Expressway Proximity on Exports by Distance Band

	ln(amount)			
	(1)	(2)	(3)	(4)
Distance _{it} (0-10km)	0.090*** (0.025)	0.102*** (0.029)	-0.102* (0.054)	-0.205** (0.093)
Distance _{it} (10-20km)	0.055** (0.024)	0.070*** (0.027)	-0.130** (0.055)	-0.159 (0.099)
Distance _{it} (20-30km)	0.064*** (0.024)	0.068** (0.027)	0.006 (0.056)	-0.045 (0.109)
Distance _{it} (30-40km)	0.048* (0.025)	0.052* (0.027)	-0.045 (0.059)	-0.117 (0.129)
Distance _{it} (40-50km)	0.035 (0.028)	0.038 (0.031)	-0.066 (0.067)	0.005 (0.108)
Distance _{it} (50-60km)	-0.037 (0.027)	-0.045 (0.030)	0.016 (0.073)	0.111 (0.199)
Constant	7.302*** (0.022)	7.295*** (0.026)	7.554*** (0.047)	7.336*** (0.083)
Firm-Product FE	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes
Region		Eastern	Central	Western
Observations	35,786,072	32,836,500	1,689,358	1,260,214
Adjusted R ²	0.669	0.661	0.771	0.7232

Notes: This table reports estimates from equation (17), which captures the distance-gradient effects of expressway proximity on firm-level exports. The dependent variable is the logarithm of export amounts. Firms are classified into 10km distance bands based on their annual distance to the nearest expressway. The omitted reference group consists of firms located between 60 and 100km from the network. Column (1) reports results for the full sample, while Columns (2)-(4) present results separately for Eastern, Central, and Western China. All specifications include firm-product and country-year fixed effects. Robust standard errors are clustered at the firm level and reported in parentheses. Statistical significance is indicated by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table 3: Average Effects of Expressway Proximity on Exports

	ln(amount)			
	(1)	(2)	(3)	(4)
Panel A: 40km threshold				
Treat _{it}	0.057*** (0.015)	0.063*** (0.017)	-0.070* (0.040)	-0.162** (0.080)
Constant	7.322*** (0.015)	7.320*** (0.016)	7.525*** (0.037)	7.311*** (0.078)
Region		Eastern	Central	Western
Firm-Product FE	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes
Observations	35,786,072	32,836,500	1,689,358	1,260,214
Adjusted R ²	0.669	0.661	0.771	0.722
Panel B: County intersection				
Treat _{it} (County Intersected by Expressway)	0.031** (0.014)	0.042*** (0.015)	-0.129*** (0.035)	-0.046 (0.050)
Constant	7.356*** (0.009)	7.353*** (0.010)	7.538*** (0.021)	7.183*** (0.032)
Region		Eastern	Central	Western
Firm-Product FE	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes
Observations	35,930,382	32,836,500	1,689,358	1,260,214
Adjusted R ²	0.669	0.661	0.771	0.722
Panel C: Continuous				
ln(Distance to Expressway)	-0.029*** (0.007)	-0.033*** (0.008)	0.023 (0.015)	0.059*** (0.023)
Constant	7.629*** (0.062)	7.668*** (0.068)	7.252*** (0.136)	6.644*** (0.196)
Region		Eastern	Central	Western
Firm-Product FE	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes
Observations	35,930,382	32,836,500	1,689,358	1,260,214
Adjusted R ²	0.669	0.661	0.771	0.722

Notes: This table reports the average effects of the ENE on firm exports under alternative treatment definitions. Panel A defines treatment using a 40km proximity threshold to the expressway network. Column (1) includes the full sample, while Columns (2)–(4) report results separately for Eastern, Central, and Western China. Panel B defines treatment at the county level based on whether the county is intersected by the expressway network in a given year. Panel C uses a continuous measure of exposure, defined as the logarithm of the distance from the firm to the expressway network, and reports the implied elasticity of exports with respect to distance. All specifications include firm–product and country–year fixed effects. Standard errors are clustered at the firm level and reported in parentheses. *** Statistically significant at the 1% level; ** at the 5% level; and * at the 10% level.

Table 4: Heterogeneous Effects of Expressway Proximity on Exports

	ln(amount)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: By Sector and Ownership							
Treat _{it}	0.056*** (0.016)	0.112 (0.098)	0.089*** (0.026)	0.030* (0.017)	0.163*** (0.044)	0.009 (0.025)	0.046*** (0.016)
Group	Manufacturing	Non-Manufacturing	High Road Reliance	Low Road Reliance	SOEs	FFs	PFs
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	35,410,072	376,000	14,214,195	21,571,877	4,894,046	15,054,912	15,837,114
Adjusted R ²	0.667	0.767	0.656	0.741	0.664	0.658	0.690
Panel B: By Country, Product Type, and Trade Regime							
Treat _{it}	0.062*** (0.020)	0.039** (0.017)	0.043 (0.043)	0.008 (0.031)	0.071*** (0.017)	0.183*** (0.069)	0.017 (0.018)
Group	OECD	Non-OECD	Capital	Intermediate	Final	Processing	Ordinary
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	17,539,390	18,246,682	5,339,339	5,071,370	25,375,363	2,972,511	8,610,578
Adjusted R ²	0.643	0.726	0.728	0.633	0.614	0.639	0.651

Notes: This table presents the heterogeneous effects of the ENE on exports across different industries, ownership structures, countries, product types, and trade regimes. Each column reports a separate regression based on equation (18), with the dependent variable being the logarithm of export amounts. In Panel A, Columns (1)–(2) distinguish between manufacturing and non-manufacturing industries; Columns (3)–(4) compare industries with high versus low road reliance, categorized by their transportation input share; and Columns (5)–(7) consider variations by ownership type (state-owned enterprises, foreign firms, and private firms). In Panel B, Columns (1)–(2) report the effects on exports to OECD versus non-OECD countries; Columns (3)–(5) assess exports of capital goods, intermediate goods, and final goods; and Columns (6)–(7) examine firms engaged in processing trade versus ordinary trade. Robust standard errors are clustered at the firm level and reported in parentheses. Statistical significance is indicated by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table 5: Instrumental Variable Estimates of Expressway Effects on Exports

	ln(amount)			
	(1)	(2)	(3)	(4)
Panel A: Least Cost Path Spanning Tree Network				
Treat _{it}	0.454*** (0.139)	0.553*** (0.155)	-0.621* (0.321)	-0.226 (0.227)
Firm-Product FE	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes
Region		Eastern	Central	Central
Observations	35,786,072	32,836,500	1,689,358	1,260,214
Kleibergen-Paap F-statistic	208.586	163.886	56.662	41.957
Panel B: Least Cost Path Spanning Tree Network Excluding Target Cities				
Treat _{it}	0.609*** (0.207)	0.582** (0.261)	-1.120 (0.987)	-0.180 (0.187)
Firm-Product FE	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes
Region		Eastern	Central	Central
Target Cities	Drop	Drop	Drop	Drop
Observations	10,028,479	9,526,461	372,107	129,911
Kleibergen-Paap F-statistic	73.110	42.541	10.358	32.802
Panel C: Ming Dynasty Courier Road (1368 - 1644)				
Treat _{it}	0.645** (0.272)	0.814** (0.370)	-0.572 (0.365)	-0.365 (0.238)
Firm-Product FE	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes
Region		Eastern	Central	Central
Observations	35,547,870	32,833,420	1,516,957	1,197,493
Kleibergen-Paap F-statistic	76.915	46.699	26.670	40.174

Notes: This table addresses potential endogeneity concerns using instrumental variable estimation. Panel A uses the least-cost path spanning tree network as an instrument for the expressway expansion measure. Panel B employs the same instrument but restricts the sample to non-target cities. Panel C uses the Ming Dynasty courier road network as an alternative instrument. Following [Banerjee et al. \(2020\)](#), firms located in Inner Mongolia, Jilin, Heilongjiang, Tibet, Qinghai, and Xinjiang are excluded in Panel C because these regions were not covered by the Ming Dynasty road network. The endogenous regressor is the interaction between an indicator for whether the firm's county is traversed by the network and the city-level budget ratio. The dependent variable is the logarithm of export amounts. Column (1) reports results for the full sample, while Columns (2)–(4) report results separately for Eastern, Central, and Western China. All specifications include firm–product and country–year fixed effects. Standard errors are clustered at the firm level and reported in parentheses. The first-stage F-statistic is reported at the bottom of each panel. Statistical significance is denoted by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table 6: Robustness of Baseline Export Effects

	ln(amount)				
	(1)	(2)	(3)	(4)	(5)
Treat _{it}	0.067*** (0.017)	0.053*** (0.016)	0.068*** (0.022)	-0.054 (0.056)	0.075** (0.033)
Constant	7.579*** (0.016)	7.339*** (0.015)	7.757*** (0.020)	7.456*** (0.054)	7.493*** (0.031)
Observations	24,389,063	32,625,435	9,100,283	42,400,442	7,407,857
Adjusted R ²	0.650	0.666	0.652	0.607	0.635
Group	Excl. Railway Hubs	Excl. Inland Ports	Excl. Urban Districts	Intermediary Only	Balanced Panel

Notes: This table reports robustness checks for the baseline result in Table 3, Column (1). Column (1) excludes firms in railway hub cities; Column (2) excludes firms in inland port cities; Column (3) excludes firms in urban districts (*shi xia qu*) where expressway access is convenient. Column (4) examines the effect on intermediary (trading) firms, which were excluded from the baseline sample. Column (5) uses a balanced panel of firms continuously active from 2001 to 2013. All specifications are based on equation (18) and include firm–product and country–year fixed effects. Robust standard errors are clustered at the firm level. Statistical significance is denoted *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table 7: Effects of Expressway Proximity on Export Prices

	ln(price)			
	(1)	(2)	(3)	(4)
Distance _{it} (0-10km)	-0.032** (0.015)	-0.048*** (0.016)	0.085** (0.039)	0.053 (0.063)
Distance _{it} (10-20km)	-0.021 (0.015)	-0.038** (0.016)	0.090** (0.041)	0.040 (0.077)
Distance _{it} (20-30km)	0.018 (0.015)	0.014 (0.016)	0.023 (0.037)	0.055 (0.054)
Distance _{it} (30-40km)	0.021 (0.016)	0.013 (0.017)	0.056* (0.034)	0.067 (0.078)
Distance _{it} (40-50km)	-0.002 (0.016)	-0.000 (0.017)	-0.048 (0.050)	-0.121 (0.085)
Distance _{it} (50-60km)	0.036** (0.015)	0.032* (0.017)	0.045 (0.040)	-0.032 (0.064)
Firm-Product FE	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes
Region		Eastern	Central	Western
Observations	35,786,072	32,836,500	1,689,358	1,260,214
Adjusted R ²	0.897	0.894	0.926	0.902

Notes: Building on Table 2, this table examines the effects of the ENE on export prices for firms located at different distances from the network. The dependent variable is the logarithm of FOB (free-on-board) export unit prices, which capture domestic transportation costs and port-related charges but exclude international shipping costs. The treated group includes firms located within 40km of the expressway network in each year, whereas the control group comprises firms located beyond 40km. Column (1) reports the results for the full sample, while Columns (2)-(4) present findings for firms in Eastern, Central, and Western China, respectively. All regressions incorporate firm-product and country-year fixed effects, and robust standard errors are clustered at the firm level and are reported in parentheses. Statistical significance is denoted by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table 8: Effects of Expressway Proximity on Extensive Margins of Trade

	Entry				Exit			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Treat_{it}$	-0.036*** (0.004)	-0.045*** (0.005)	0.021** (0.010)	0.009 (0.029)	0.006** (0.003)	-0.004 (0.003)	0.017** (0.008)	0.046 (0.029)
Constant	0.220*** (0.004)	0.225*** (0.004)	0.193*** (0.009)	0.218*** (0.026)	0.098*** (0.003)	0.104*** (0.003)	0.113*** (0.007)	0.113*** (0.025)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region		Eastern	Central	Western		Eastern	Central	Western
Observations	1,530,914	1,381,567	103,076	46,271	1,369,443	1,236,542	91,313	41,588
Adjusted R ²	0.201	0.200	0.213	0.204	0.161	0.160	0.167	0.158
	ln(Number of Products)				ln(Number of Destination Countries)			
	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
$Treat_{it}$	0.043*** (0.009)	0.079*** (0.011)	-0.064*** (0.019)	-0.134** (0.053)	0.064*** (0.010)	0.096*** (0.012)	-0.030 (0.020)	-0.113** (0.055)
Constant	1.272*** (0.008)	1.261*** (0.010)	1.110*** (0.017)	1.295*** (0.047)	1.299*** (0.009)	1.285*** (0.011)	1.262*** (0.017)	1.247*** (0.049)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region		Eastern	Central	Western		Eastern	Central	Western
Observations	1,545,354	1,394,990	103,729	46,635	1,545,354	1,394,990	103,729	46,635
Adjusted R ²	0.713	0.714	0.702	0.682	0.738	0.740	0.719	0.685

Notes: This table examines the effects of expressway access on the extensive margins of trade using firm-year level data from Sample A. The dependent variables are: a binary indicator for market entry (Columns 1–4), a binary indicator for market exit (Columns 5–8), the logarithm of the number of HS 8-digit products exported (Columns 9–12), and the logarithm of the number of destination countries (Columns 13–16). All specifications are based on equation (18) and include firm and year fixed effects. Robust standard errors are clustered at the firm level and reported in parentheses. Statistical significance is denoted by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table 9: Effects of Expressway Proximity on Market Concentration and Resource Misallocation

	At the County-Year Level				At the County-Industry-Year Level			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Dependent Variable - $\ln(\text{HHI})$								
Treat _{it} (County)	-0.070*** (0.004)	-0.051*** (0.005)	-0.075*** (0.006)	-0.092*** (0.010)	-0.063*** (0.004)	-0.073*** (0.008)	-0.054*** (0.005)	-0.053*** (0.009)
Constant	0.227*** (0.003)	0.120*** (0.004)	0.224*** (0.005)	0.370*** (0.007)	0.633*** (0.003)	0.560*** (0.006)	0.694*** (0.004)	0.729*** (0.006)
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes				
Industry-Year FE					Yes	Yes	Yes	Yes
Region		Eastern	Central	Western		Eastern	Central	Western
Observations	37,433	12,500	15,328	9,605	440,139	213,463	159,661	67,015
Adjusted R ²	0.447	0.365	0.316	0.439	0.244	0.255	0.173	0.171
Panel B: Dependent Variable - $\ln(\text{Misallocation})$								
Treat _{it} (County)	0.015* (0.008)	-0.024* (0.014)	0.035*** (0.011)	0.046*** (0.015)	0.004 (0.006)	-0.012 (0.009)	0.026** (0.010)	0.014 (0.016)
Constant	0.661*** (0.005)	0.708*** (0.010)	0.657*** (0.007)	0.604*** (0.010)	0.613*** (0.005)	0.637*** (0.007)	0.588*** (0.007)	0.576*** (0.012)
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes				
Industry-Year FE					Yes	Yes	Yes	Yes
Region		Eastern	Central	Western		Eastern	Central	Western
Observations	35,723	12,368	14,871	8,484	278,281	153,995	89,620	34,666
Adjusted R ²	0.122	0.193	0.098	0.088	0.041	0.050	0.033	0.039

Notes: This table analyzes the effects of the ENE on market competition and resource misallocation. The dependent variables in Panel A are the Herfindahl-Hirschman Index (HHI), while those in Panel B represent misallocation, measured by the variance of $\ln(\text{TFPR})$. Data in Columns (1)-(4) is aggregated at the county-year level, while data in Columns (5)-(8) is aggregated at the county-industry-year level. Columns (1) and (5) present results for all regions, whereas the remaining columns provide results for firms in Eastern, Central, and Western China, respectively. The regressions in Columns (1)-(4) include city and year fixed effects, while those in Columns (5)-(8) include city and industry-year fixed effects. Robust standard errors, clustered at the county level, are reported in parentheses. Statistical significance is denoted by *** at the 1% level, ** at the 5% level, and * at the 10% level.

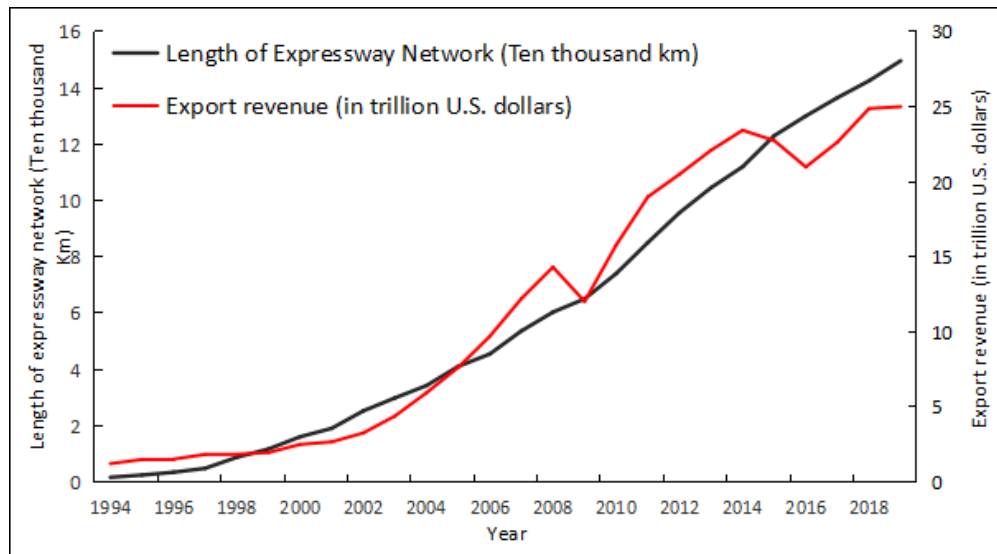
Online Appendix

Expressway Expansion, Export Competition, and Spatial Divergence

The online appendix shows additional figures, tables, and supplementary that are omitted from the main paper due to space constraints.

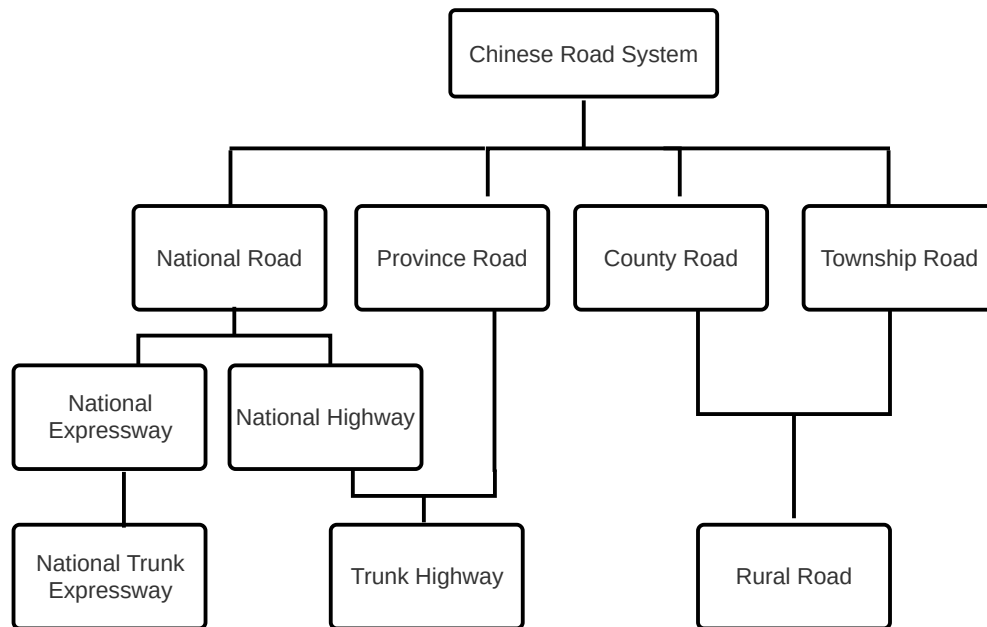
Appendix A: Other figures

Figure A1: Development of Chinese Expressways and Exports



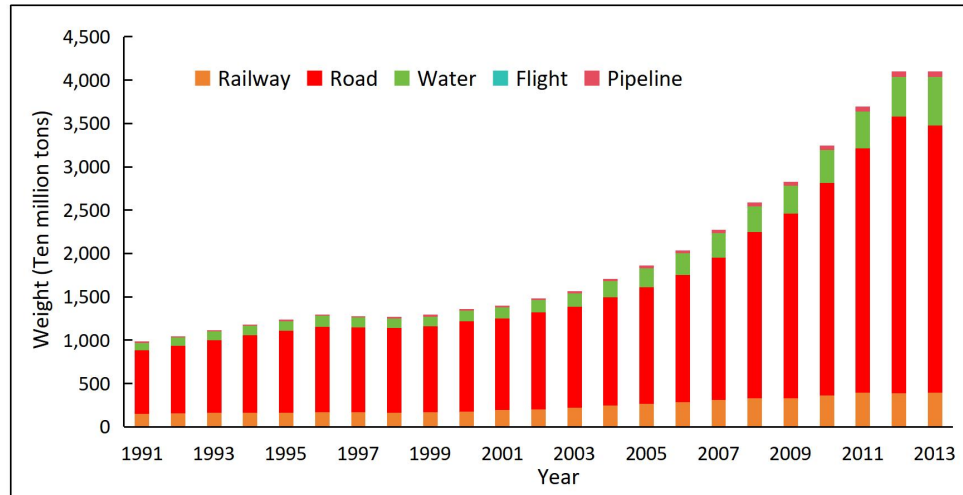
Note: The data are from The Chinese Statistical Yearbooks. The black line is the cumulative total length of the expressway network each year. The red line is the Chinese export revenue each year.

Figure A2: Structure of China's Road Classification System



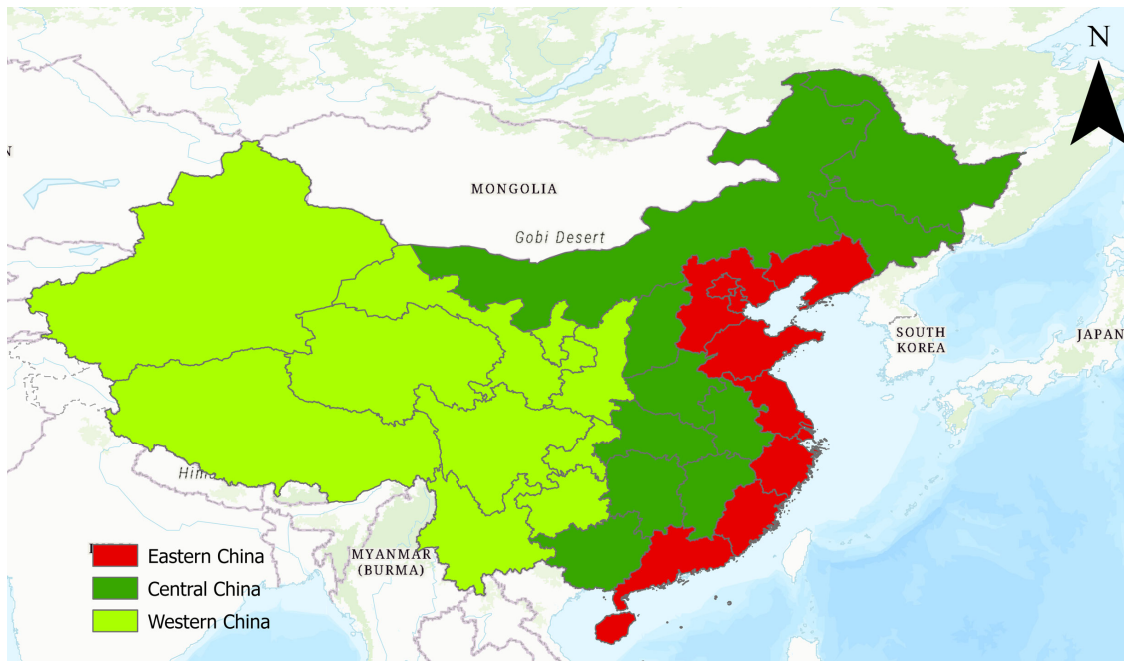
Notes: This figure shows the Chinese road system. The Chinese road system includes national, province, county, and township roads. Each type of road is designed to meet specific transportation needs, from high-speed travel between major cities to local access in rural areas. The main differences between these roads are their width, quality, speed, and the areas they connect. Also, expressways charge tolls.

Figure A3: Proportion of Different Types of Cargo Transportation in China



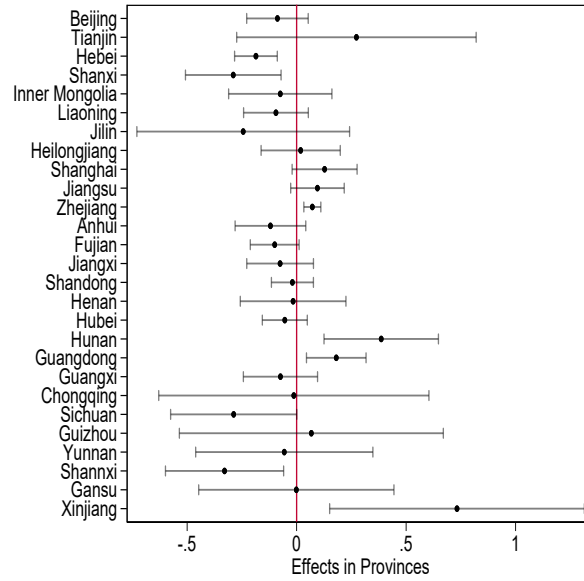
Notes: This figure presents cargo transportation volumes by different modes from 1991 to 2013, using data from the China Transportation Yearbook.

Figure A4: Regional Classification of Eastern, Central, and Western China



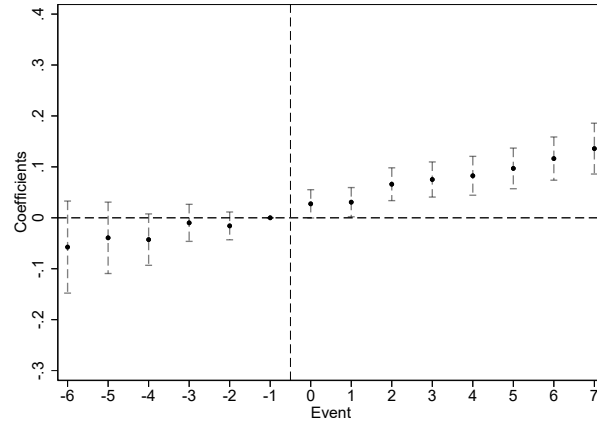
Notes: The figure shows the geographical distribution of Eastern, Central, and Western China.

Figure A5: Differential Effects Across Provinces

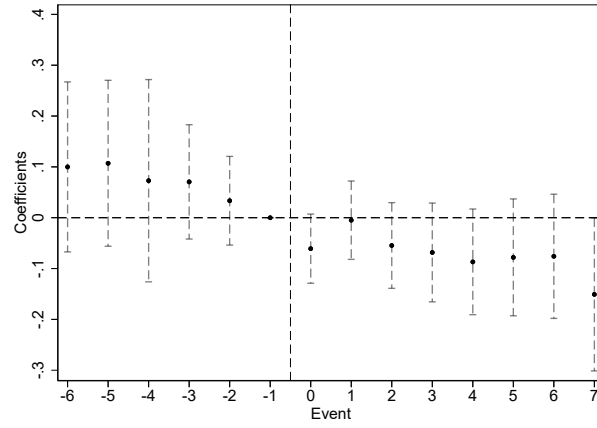


Notes: This figure illustrates the impact of the ENE on firms' exports across provinces, using a 40km threshold to define the treated group. The black dots represent estimated coefficients for each province, while the error bars indicate 95% confidence intervals. The regressions control for firm-product and country-year fixed effects, with robust standard errors clustered at the firm level.

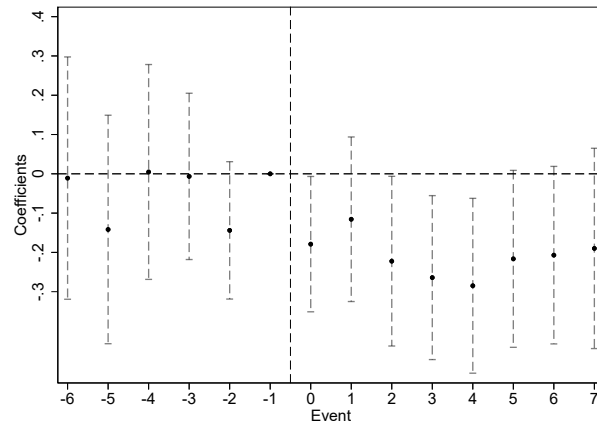
Figure A6: Dynamic Effects on Exports



(a) Eastern China



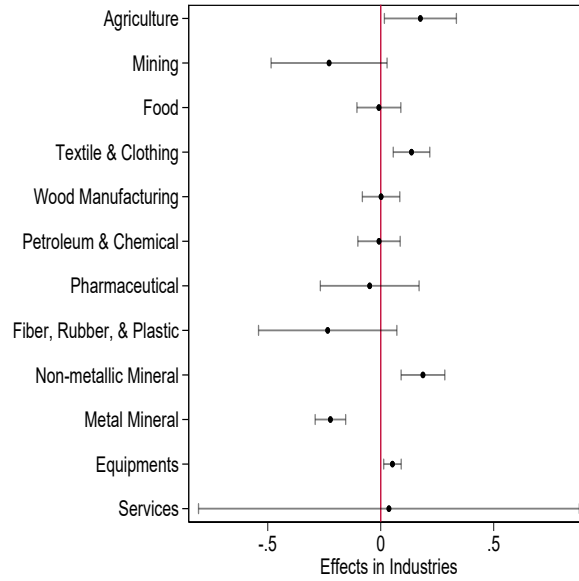
(b) Central China



(c) Western China

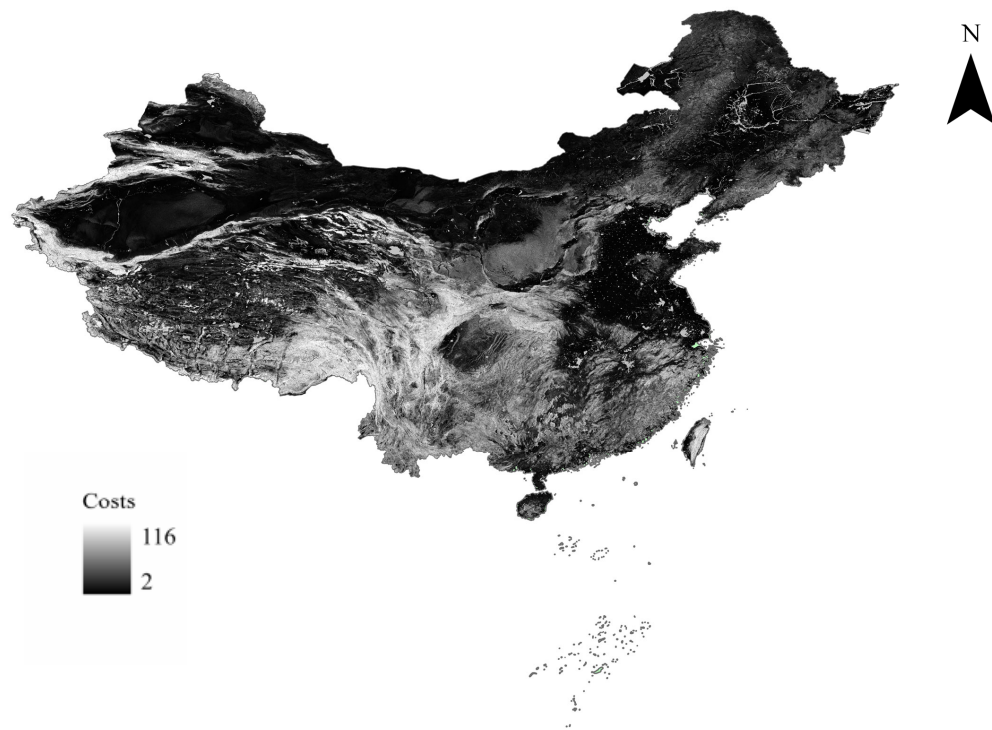
Notes: The figure shows the dynamic effects of the ENE on exports by estimating equation (19). The black dots represent the changes in the effects on export amounts over time for the Eastern, Central, and Western regions of China. The upper and lower bounds represent 95% confidence intervals. The regressions also control for firm-product and country-year fixed effects, and the robust standard errors are clustered at the firm level.

Figure A7: Differential Effects Across Industries



Notes: This figure illustrates the impact of the ENE on firms' exports across industries, using a 40km threshold to define the treated group. The black dots represent estimated coefficients for each industry, while the error bars indicate 95% confidence intervals. The regressions control for firm-product and country-year fixed effects, with robust standard errors clustered at the firm level. Industries are classified based on the 2002 Chinese Industry Classification (CIC) system: Agriculture (CIC code < 6), Mining (CIC codes 6-12), Food Manufacturing (CIC codes 13-15), Textiles and Clothing (CIC codes 17-19), Wood Manufacturing (CIC codes 20-24), Petroleum and Chemical (CIC codes 25, 26), Pharmaceuticals (CIC codes 7, 27), Fiber, Rubber, and Plastics (CIC codes 8, 28-29), Non-metallic Minerals (CIC codes 9, 30), Metal Minerals (CIC codes 10, 31-33), Equipment Manufacturing (CIC codes 34-43), and Services (CIC codes ≥ 62). Other CIC industries are excluded due to limited observations.

Figure A8: Construction Cost Grid



Notes: The figure illustrates the construction cost grid used as input for the least-cost path algorithm. The color gradient ranges from white to black, from high traversal costs to low per 900 square meters. Following [Faber \(2014\)](#), cost values are assigned based on factors such as terrain gradient (slope) and land cover types, including features like water bodies, wetlands, and urbanized areas.

Appendix B: Other tables

Table B1: Chinese Transportation Atlases (1992 - 2013)

Year	Title (English)	Publisher	Edition number	Print number
2013	China Transport and Tourism Atlas	China Communications Press	Version 2, January 2014	1st printing, January 2014
2012	Newly compiled China Transport and Tourism Atlas	Shandong Provincial Map Publishing House	Version 6, January 2013	6th printing, January 2013
2011	China Transportation Atlas	SinoMaps Press	Version 1, January 2012	1st printing, January 2012
2010	China Transportation Atlas	Shandong Provincial Map Publishing House	Version 5, January 2011	9th printing, January 2011
2010	China Transportation Atlas	SinoMaps Press	Version 1, January 2009	8th printing, January 2011
2009	China Transport and Tourism Atlas	China Tourism Publishing House	Version 4, February 2010	8th printing, February 2010
2009	China Transportation Atlas	SinoMaps Press	Version 1, January 2009	4th printing, January 2010
2008	China Transportation Atlas	SinoMaps Press	Version 1, January 2009	2nd printing, April 2009
2007	China Transport and Tourism Atlas	Shandong Provincial Map Publishing House	Version 4, January 2008	11th printing, April 2008
2006	China Transportation Atlas	Fujian Map Publishing House	Version 1, January 2007	1st printing, January 2007
2006	China Transportation Atlas	SinoMaps Press	Version 2, March 2007	4th printing, April 2007
2006	China Transportation Atlas	SinoMaps Press	Version 1, January 2007	2nd printing, January 2007
2005	China Transport and Tourism Atlas	China Communications Press	Version 1, April 2006	5th printing, April 2006
2005	China Transport and Tourism Atlas	SinoMaps Press	Version 1, January 2006	2nd printing, January 2006
2004	China Transportation Atlas	China Communications Press	Version 2, January 2005	7th printing, January 2005
2003	China Transportation Atlas	China Communications Press	Version 1, October 2004	6th printing, October 2004
2003	Mini China Atlas	SinoMaps Press	Version 1, January 2004	7th printing, January 2004
2002	China Transportation Atlas	SinoMaps Press	Version 2, June 2003	12th printing, June 2003
2001	China Transportation Atlas	SinoMaps Press	Version 3, July 2002	10th printing, July 2002
2000	China Transportation Atlas	Hunan Map Publishing House	Version 1, January 2001	2nd printing, April 2001
1999	China Traffic Atlas	SinoMaps Press	Version 1, July 2000	2nd printing, July 2000
1999	Newly China Traffic Atlas	Academy Press	Version 3, February 2000	3rd printing, February 2000
1998	China Transportation Atlas	Xi'an Map Publishing House	Version 3, May 1999	3rd printing, May 1999
1997	Newly China Traffic Atlas	SinoMaps Press	Version 1, March 1998	1st printing, March 1998
1996	China Traffic Atlas	Surveying and Mapping Publishing House	Version 1, January 1997	1st printing, January 1997
1995	Newly China Traffic Atlas	Geological Publishing House	Version 1, January 1996	1st printing, January 1996
1994	China Transport and Tourism Atlas	SinoMaps Press	Version 1, March 1995	7th printing, March 1995
1993	China Transportation Atlas	SinoMaps Press	Version 5, February 1994	46th printing, March 1995
1992	China Transportation Atlas	SinoMaps Press	Version 5, February 1993	45th printing, February 1993

Notes: This table shows the atlases that were used to hand-collect and digitize the expressway information each year.

Table B2: Summary Statistics for the Industrial Firm Sample

	Group	2000-2013			2000			2013		
		Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.
Distance to Nearest Expressway (km)	All	2,410,761	15.473	16.271	72,537	23.201	22.990	253,212	13.706	14.124
	Eastern	1,790,813	14.579	14.705	54,636	21.957	21.648	169,572	13.005	12.627
	Central	454,131	18.409	19.471	11,998	29.481	26.216	62,987	15.307	16.372
	Western	165,817	17.089	20.990	5,903	21.947	25.781	20,653	14.582	17.507
	SOEs	71,970	17.224	20.345	10,007	23.941	24.953	2,069	12.618	14.171
	FFs	541,051	12.942	14.050	19,015	17.040	18.015	42,110	11.653	12.440
	PFs	1,797,740	16.186	16.651	43,515	25.723	23.938	209,033	14.131	14.403
ln(HHI)	All	37,433	0.200	0.239	2,756	0.277	0.268	2,736	0.134	0.201
	Eastern	12,500	0.096	0.145	918	0.149	0.182	889	0.069	0.141
	Central	15,328	0.197	0.211	1,129	0.295	0.254	1,099	0.123	0.184
	Western	9,605	0.339	0.301	709	0.414	0.306	748	0.228	0.247
ln(misallocation)	All	35,723	0.667	0.393	2,558	0.582	0.370	2,661	0.702	0.350
	Eastern	12,368	0.696	0.333	899	0.627	0.280	879	0.716	0.265
	Central	14,871	0.670	0.404	1,055	0.575	0.356	1,073	0.721	0.375
	Western	8,484	0.620	0.446	604	0.528	0.486	709	0.659	0.398

Notes: This table reports summary statistics for Sample B (industrial firms from the Annual Survey of Industrial Firms), spanning 2000–2013, with separate snapshots for 2000 and 2013. The unit of observation is firm-year for the distance to the nearest expressway and county-year for aggregated variables (HHI, misallocation). Subgroup statistics are reported by region (Eastern, Central, and Western China) and by ownership type (state-owned enterprises, foreign firms, and private firms).

Table B3: Effects of Expressway Proximity on Exports by Distance Band With Different Controls

	ln(amount)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Full Sample and Eastern China								
Distance _{it} (0-10km)	0.061*** (0.022)	0.080*** (0.025)	0.085** (0.036)	0.094** (0.039)	0.076*** (0.025)	0.101*** (0.028)	0.094*** (0.031)	0.096*** (0.033)
Distance _{it} (10-20km)	0.048** (0.021)	0.066*** (0.023)	0.064* (0.034)	0.071** (0.036)	0.067*** (0.023)	0.091*** (0.026)	0.070** (0.029)	0.071** (0.031)
Distance _{it} (20-30km)	0.066*** (0.021)	0.082*** (0.023)	0.066** (0.032)	0.069** (0.034)	0.075*** (0.024)	0.100*** (0.026)	0.059** (0.028)	0.067** (0.029)
Distance _{it} (30-40km)	0.034 (0.023)	0.051** (0.025)	0.047 (0.035)	0.051 (0.038)	0.044* (0.026)	0.068** (0.028)	0.045 (0.030)	0.051 (0.032)
Distance _{it} (40-50km)	-0.003 (0.026)	0.011 (0.027)	-0.000 (0.037)	-0.000 (0.039)	0.005 (0.029)	0.026 (0.030)	-0.009 (0.031)	-0.009 (0.033)
Distance _{it} (50-60km)	-0.041 (0.025)	-0.032 (0.026)	-0.040 (0.039)	-0.041 (0.040)	-0.042 (0.028)	-0.026 (0.030)	-0.051 (0.032)	-0.060* (0.034)
Firm-Product FE	Yes	Yes			Yes	Yes		
Firm-Product-Country FE			Yes	Yes			Yes	Yes
Product-Year FE	Yes		Yes		Yes		Yes	
Country-Year FE	Yes		Yes		Yes		Yes	
Product-Country-Year FE		Yes		Yes		Yes		Yes
Region					Eastern	Eastern	Eastern	Eastern
Observations	35,786,072	35,786,072	35,786,072	35,786,072	32,836,500	32,836,500	32,836,500	32,836,500
Adjusted R ²	0.673	0.682	0.796	0.794	0.665	0.674	0.795	0.783
Panel B: Central China and Western China								
Distance _{it} (0-10km)	-0.039 (0.043)	-0.073 (0.063)	0.001 (0.042)	-0.011 (0.072)	-0.078 (0.112)	-0.232 (0.185)	-0.258** (0.126)	-0.192 (0.225)
Distance _{it} (10-20km)	-0.099** (0.044)	-0.143** (0.065)	-0.029 (0.043)	-0.028 (0.073)	-0.040 (0.109)	-0.144 (0.179)	-0.130 (0.118)	-0.067 (0.215)
Distance _{it} (20-30km)	-0.001 (0.049)	-0.073 (0.069)	0.084* (0.047)	0.062 (0.075)	0.086 (0.112)	0.076 (0.204)	0.116 (0.141)	0.170 (0.242)
Distance _{it} (30-40km)	-0.013 (0.057)	-0.083 (0.083)	0.065 (0.058)	0.055 (0.093)	-0.006 (0.131)	-0.048 (0.193)	-0.083 (0.150)	-0.173 (0.235)
Distance _{it} (40-50km)	-0.027 (0.053)	-0.051 (0.078)	0.023 (0.061)	-0.020 (0.092)	0.047 (0.103)	0.089 (0.194)	0.030 (0.107)	0.055 (0.211)
Distance _{it} (50-60km)	0.002 (0.053)	-0.040 (0.075)	0.057 (0.048)	0.046 (0.083)	0.051 (0.168)	0.013 (0.172)	0.092 (0.112)	0.096 (0.179)
Firm-Product FE	Yes	Yes			Yes	Yes		
Firm-Product-Country FE			Yes	Yes			Yes	Yes
Product-Year FE	Yes		Yes		Yes		Yes	
Country-Year FE	Yes		Yes		Yes		Yes	
Product-Country-Year FE		Yes		Yes		Yes		Yes
Region	Central	Central	Central	Central	Western	Western	Western	Western
Observations	1,689,358	1,689,358	1,689,358	1,689,358	1,260,214	1,260,214	1,260,214	1,260,214
Adjusted R ²	0.773	0.750	0.859	1.473	0.721	0.713	0.778	1.096

Notes: This table reports robustness checks for the distance-gradient effects presented in Table 2, incorporating progressively more demanding fixed effects structures. The dependent variable is the logarithm of export amounts. Panel A reports results for the full sample and Eastern China, while Panel B reports results for Central and Western China. Columns (1) and (5) include firm-product, product-year, and country-year fixed effects; the addition of product-year fixed effects captures product-specific preference trends and global demand shifts that may evolve differently across products over time. Columns (2) and (6) include firm-product and product-country-year fixed effects, where the latter absorbs country-specific product preference trends, including destination-product-level demand shocks, bilateral tariff changes, and exchange rate movements in each year. Columns (3) and (7) include firm-product-country, product-year, and country-year fixed effects; the firm-product-country fixed effects control for all time-invariant characteristics of a given firm-product-destination triplet, such as established buyer-supplier relationships, destination-specific product quality, and bilateral trade preferences, thereby exploiting only within-relationship variation over time. Columns (4) and (8) adopt the most demanding specification by including firm-product-country and product-country-year fixed effects, which simultaneously absorb all time-invariant relationship characteristics and all destination-product-specific demand shocks in each year. Identification in this specification relies solely on variation across firms exporting the same product to the same destination in the same year. The omitted reference group consists of firms located more than 60km from the expressway network. Robust standard errors are clustered at the firm level and reported in parentheses. Statistical significance is indicated by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table B4: Heterogeneous Effects of Expressway Proximity on Exports With a Triple-Difference

	ln(amount)					
	(1)	(2)	(3)	(4)	(5)	(6)
$Treat_{it}$	0.180* (0.093)	0.093*** (0.016)	0.002 (0.024)	0.086*** (0.026)	0.058*** (0.017)	0.021 (0.020)
$Treat_{it} * \text{Manufacturing Firm}$	-0.125 (0.094)					
$Treat_{it} * \text{High Road Reliance}$		-0.100*** (0.027)				
$Treat_{it} * \text{Private Firm}$			0.123*** (0.027)			
$Treat_{it} * \text{OECD Country}$				-0.003 (0.020)		
$Treat_{it} * \text{Final Goods}$					-0.038 (0.027)	
$Treat_{it} * \text{Processing Trade}$						0.177*** (0.040)
Constant	7.323*** (0.015)	-0.100*** (0.027)	7.323*** (0.015)	7.322*** (0.015)	7.320*** (0.015)	7.509*** (0.019)
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	35,786,072	35,786,072	35,786,072	35,786,072	35,786,072	11,583,089
Adjusted R ²	0.669	0.669	0.669	0.669	0.669	0.634

Notes: This table reports the results from a triple-difference (DDD) specification corresponding to the heterogeneous effects presented in Table 4. Each column augments the baseline difference-in-differences model by interacting the treatment variable ($Treat_{it}$) with a binary indicator variable that captures a specific dimension of heterogeneity. Column (1) interacts treatment with a manufacturing sector dummy; Column (2) with an indicator for high transportation input reliance; Column (3) with a private firm indicator; Column (4) with an OECD destination country indicator; Column (5) with a final goods indicator; and Column (6) with a processing trade indicator. The dependent variable is the logarithm of export amounts. All regressions include firm-product and country-year fixed effects. Robust standard errors are clustered at the firm level and reported in parentheses. Statistical significance is denoted by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table B5: Heterogeneous Effects of Expressway Proximity on Exports of Manufacturing and Non-manufacturing

	ln(amount)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Manufacturing Products						
Treat _{it}	0.061*** (0.017)	-0.067* (0.041)	-0.145* (0.083)	0.165*** (0.044)	0.008 (0.026)	0.044*** (0.016)
Constant	7.310*** (0.016)	7.481*** (0.038)	7.264*** (0.081)	7.316*** (0.043)	7.274*** (0.024)	7.361*** (0.015)
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Group	Eastern	Central	Western	SOEs	FFs	PFs
Observations	32,518,574	1,658,071	1,233,427	4,838,061	14,894,690	15,677,321
Adjusted R ²	0.659	0.770	0.722	0.662	0.656	0.689
Panel B: Non-manufacturing Products						
Treat _{it}	0.194* (0.108)	-0.112 (0.130)	-0.633*** (0.181)	-0.050 (0.255)	0.059 (0.103)	0.196 (0.169)
Constant	8.485*** (0.102)	9.780*** (0.116)	9.155*** (0.168)	9.068*** (0.247)	8.462*** (0.097)	8.698*** (0.156)
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Group	Eastern	Central	Western	SOEs	FFs	PFs
Observations	317,926	31,287	26,787	55,985	160,222	159,793
Adjusted R ²	0.773	0.745	0.698	0.727	0.789	0.754

Notes: This table presents the heterogeneous effects of the ENE on exports from manufacturing and non-manufacturing industries across different regions and ownership types. Each panel column represents separate regression estimates based on equation (18), with the dependent variable being the logarithm of exports. Panel A covers manufacturing industries, while Panel B covers non-manufacturing industries. Columns (1)-(3) focus on firms' exports in Eastern, Central, and Western China, respectively, whereas Columns (4)-(6) examine the exports of state-owned enterprises (SOEs), foreign firms, and private firms. Robust standard errors, clustered at the firm level, are reported in parentheses. Statistical significance is denoted by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table B6: Heterogeneous Effects of Expressway Proximity on Exports by Industry Road Reliance

	ln(amount)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: High Road Reliance Industries						
Treat _{it}	0.088*** (0.029)	0.008 (0.056)	-0.123 (0.122)	0.249*** (0.060)	0.052 (0.043)	0.074*** (0.025)
Constant	7.919*** (0.027)	8.635*** (0.052)	8.115*** (0.118)	8.065*** (0.058)	7.822*** (0.041)	8.030*** (0.024)
Group	Eastern	Central	Western	SOEs	FFs	PFs
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13,215,631	608,099	390,465	1,921,400	6,091,885	6,200,910
Adjusted R ²	0.576	0.661	0.664	0.525	0.594	0.595
Panel B: Low Road Reliance Industries						
Treat _{it}	0.039** (0.018)	-0.125** (0.053)	-0.135 (0.092)	0.110** (0.054)	-0.014 (0.027)	0.016 (0.019)
Constant	6.924*** (0.017)	6.913*** (0.051)	6.907*** (0.089)	6.861*** (0.053)	6.917*** (0.026)	6.964*** (0.018)
Group	Eastern	Central	Central	SOEs	FFs	PFs
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	19,620,869	1,081,259	869,749	2,972,646	8,963,027	9,636,204
Adjusted R ²	0.686	0.787	0.728	0.699	0.678	0.717

Notes: This table presents the heterogeneous effects of the ENE on exports for industries with high and less road reliance across different regions and ownership types. Each column within a panel represents a separate regression estimate based on equation (18), with the dependent variable being the logarithm of exports. Panel A focuses on high road reliance industries, while Panel B addresses less road reliance industries. Columns (1)-(3) present results for firms in Eastern, Central, and Western China, respectively, while Columns (4)-(6) provide estimates for state-owned enterprises (SOEs), foreign firms, and private firms. Robust standard errors are clustered at the firm level and reported in parentheses. Statistical significance is indicated by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table B7: Heterogeneous Effects of Expressway Proximity on Exports by Ownership

	ln(amount)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treat _{it}	0.191*** (0.044)	-0.056 (0.084)	-0.202 (0.125)	0.018 (0.027)	-0.195*** (0.072)	-0.490** (0.203)	0.048*** (0.017)	0.040 (0.044)	0.054 (0.140)
Constant	7.356*** (0.043)	7.180*** (0.079)	7.419*** (0.122)	7.275*** (0.026)	7.840*** (0.067)	6.917*** (0.196)	7.377*** (0.016)	7.485*** (0.041)	7.189*** (0.137)
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region	Eastern	Central	Western	Eastern	Central	Western	Eastern	Central	Western
Ownership	SOEs	SOEs	SOEs	FFs	FFs	FFs	PFs	PFs	PFs
Observations	4,278,774	405,298	209,974	14,521,379	399,021	134,512	14,036,347	885,039	915,728
Adjusted R ²	0.641	0.791	0.773	0.655	0.725	0.738	0.683	0.784	0.700

Notes: This table presents the heterogeneous effects of the ENE on exports by firm ownership across different regions. Each column represents a separate regression estimate based on equation (18), with the dependent variable being the logarithm of exports. Columns (1)-(3), (4)-(6), and (7)-(9) report results for state-owned enterprises (SOEs), foreign firms, and private firms, respectively, across Eastern, Central, and Western China. Robust standard errors are clustered at the firm level and reported in parentheses. Statistical significance is denoted by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table B8: Heterogeneous Effects of Expressway Proximity on Exports by Continents

	ln(amount)					
	(1)	(2)	(3)	(4)	(5)	(6)
Treat _{it}	0.059*** (0.018)	0.058** (0.023)	0.085*** (0.026)	-0.020 (0.035)	-0.032 (0.029)	-0.069* (0.042)
Constant	7.276*** (0.017)	7.347*** (0.022)	7.719*** (0.025)	7.325*** (0.034)	7.039*** (0.028)	7.039*** (0.040)
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent	Asia	Europe	North America	South America	Africa	Oceania
Observations	15,298,770	9,866,788	4,824,584	1,916,804	2,442,478	1,436,648
Adjusted R ²	0.702	0.663	0.699	0.764	0.797	0.757

Notes: This table presents the heterogeneous effects of the ENE on exports to different continents. Each panel column represents a separate regression estimate based on equation (18), with the dependent variable being the logarithm of exports. Robust standard errors, clustered at the firm level, are reported in parentheses. Statistical significance is indicated by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table B9: Heterogeneous Effects of Expressway Proximity on Exports by OECD vs. Non-OECD Countries

	ln(amount)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OECD Country						
Treat _{it}	0.068*** (0.022)	-0.037 (0.046)	-0.397*** (0.121)	0.137** (0.060)	0.005 (0.031)	0.066*** (0.020)
Constant	7.458*** (0.021)	7.950*** (0.043)	7.613*** (0.118)	7.786*** (0.059)	7.416*** (0.029)	7.478*** (0.019)
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region	Eastern	Central	Western	SOEs	FFs	PFs
Observations	16,468,756	682,623	388,011	2,246,023	8,409,307	6,884,060
Adjusted R ²	0.638	0.728	0.697	0.584	0.653	0.652
Panel B: Non-OECD Country						
Treat _{it}	0.044** (0.018)	-0.088* (0.048)	-0.068 (0.101)	0.129** (0.053)	-0.003 (0.027)	0.032 (0.020)
Constant	7.196*** (0.017)	7.234*** (0.046)	7.187*** (0.098)	7.008*** (0.051)	7.138*** (0.026)	7.293*** (0.019)
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Group	Eastern	Central	Western	SOEs	FFs	PFs
Observations	16,367,744	1,006,735	872,203	2,648,023	6,645,605	8,953,054
Adjusted R ²	0.720	0.810	0.748	0.734	0.713	0.741

Notes: This table presents the heterogeneous effects of the ENE on exports to OECD and non-OECD countries across different regions and ownership types. Each panel column represents a separate regression estimate based on equation (18), with the dependent variable being the logarithm of exports. Panel A focuses on exports to OECD countries, while Panel B focuses on exports to non-OECD countries. Columns (1)-(3) in each panel present results for firms in Eastern, Central, and Western China, respectively, while Columns (4)-(6) provide estimates for state-owned enterprises (SOEs), foreign firms, and private firms. Robust standard errors, clustered at the firm level, are reported in parentheses. Statistical significance is indicated by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table B10: Heterogeneous Effects of Expressway Proximity on Exports by Product Types

	ln(amount)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Capital Goods						
Treat _{it}	0.036 (0.046)	0.041 (0.091)	-0.035 (0.143)	0.369*** (0.077)	-0.085 (0.071)	0.077* (0.046)
Constant	5.298*** (0.044)	3.597*** (0.089)	4.705*** (0.140)	4.498*** (0.076)	5.780*** (0.069)	4.858*** (0.045)
Group	Eastern	Central	Western	SOEs	FFs	PFs
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,904,061	243,284	191,994	790,817	2,164,687	2,383,835
Adjusted R ²	0.722	0.793	0.746	0.752	0.694	0.755
Panel B: Intermediate Goods						
Treat _{it}	0.017 (0.034)	-0.137 (0.116)	-0.467*** (0.163)	0.143 (0.099)	-0.027 (0.051)	-0.011 (0.039)
Constant	7.044*** (0.032)	6.986*** (0.111)	6.916*** (0.159)	6.973*** (0.096)	7.000*** (0.049)	7.066*** (0.037)
Group	Eastern	Central	Western	SOEs	FFs	PFs
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,589,583	286,042	195,745	656,462	2,356,168	2,058,740
Adjusted R ²	0.626	0.718	0.653	0.632	0.628	0.651
Panel C: Final Goods						
Treat _{it}	0.079*** (0.018)	-0.055 (0.041)	-0.111 (0.092)	0.136*** (0.046)	0.033 (0.028)	0.058*** (0.018)
Constant	7.799*** (0.017)	8.462*** (0.038)	7.963*** (0.089)	8.039*** (0.045)	7.663*** (0.026)	7.948*** (0.017)
Group	Eastern	Central	Western	SOEs	FFs	PFs
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	23,342,856	1,160,032	872,475	3,446,767	10,534,057	11,394,539
Adjusted R ²	0.608	0.688	0.667	0.554	0.637	0.610

Notes: This table presents the heterogeneous effects of the ENE on exports of capital, intermediate, and final goods across different regions and ownership types. Each panel column represents a separate regression estimate based on equation (18), with the dependent variable being the logarithm of exports. Panels A, B, and C focus on capital goods, intermediate goods, and final goods, respectively. Columns (1)-(3) in each panel show results for firms in Eastern, Central, and Western China, while Columns (4)-(6) provide estimates for state-owned enterprises (SOEs), foreign firms, and private firms. Robust standard errors are clustered at the firm level and are reported in parentheses. Statistical significance is denoted by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table B11: Heterogeneous Effects of Expressway Proximity on Exports by Trade Regime

	ln(amount)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Processing-Trade Exports						
Treat _{it}	0.209*** (0.073)	-0.269** (0.121)	-0.333* (0.193)	0.158 (0.115)	0.187* (0.104)	0.123** (0.059)
Constant	8.056*** (0.068)	8.546*** (0.106)	8.500*** (0.184)	8.821*** (0.112)	7.920*** (0.097)	8.637*** (0.051)
Group	Eastern	Central	Western	SOEs	FFs	PFs
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,922,352	38,353	11,806	393,781	2,388,180	190,550
Adjusted R ²	0.636	0.807	0.829	0.535	0.653	0.630
Panel B: Ordinary-Trade Exports						
Treat _{it}	0.018 (0.020)	0.005 (0.044)	-0.050 (0.119)	0.106*** (0.028)	-0.015 (0.030)	0.004 (0.028)
Constant	7.590*** (0.018)	8.149*** (0.039)	7.573*** (0.115)	7.611*** (0.027)	7.416*** (0.027)	7.786*** (0.026)
Group	Eastern	Central	Western	SOEs	FFs	PFs
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,982,800	408,846	218,932	2,075,356	3,127,376	3,407,846
Adjusted R ²	0.643	0.743	0.730	0.645	0.670	0.637

Notes: This table presents the heterogeneous effects of the ENE on exports by firms engaged exclusively in processing exports, general exports, or both, across different regions and ownership types. Each panel column represents a separate regression estimate based on equation (18), with the dependent variable being the logarithm of exports. Panels A, B, and C focus on firms involved solely in processing exports, general exports, and mixed exports, respectively. Columns (1)-(3) in each panel present results for firms in Eastern, Central, and Western China, while Columns (4)-(6) provide estimates for state-owned enterprises (SOEs), foreign firms, and private firms. Missing results in Columns (3) and (4) of Panel A are due to the limited availability of observations. Robust standard errors are clustered at the firm level and are reported in parentheses. Statistical significance is denoted by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table B12: Heterogeneous Effects of Expressway Proximity on Exports by Export Types

	ln(amount)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Processing-Only Firms						
Treat _{it}	0.068 (0.299)	-0.032 (0.448)			0.083 (0.262)	1.033** (0.438)
Constant	8.390*** (0.282)	7.877*** (0.349)			8.358*** (0.247)	7.616*** (0.402)
Group	Eastern	Central			FFs	PFs
Firm-Product FE	Yes	Yes			Yes	Yes
Country-Year FE	Yes	Yes			Yes	Yes
Observations	229,143	1,415			223,247	4,562
Adjusted R ²	0.660	0.891			0.662	0.719
Panel B: Ordinary-Only Firms						
Treat _{it}	-0.011 (0.056)	0.076 (0.126)	-0.226 (0.261)	0.056 (0.093)	-0.129* (0.073)	0.141* (0.083)
Constant	7.696*** (0.053)	8.334*** (0.108)	8.194*** (0.253)	7.593*** (0.087)	8.024*** (0.066)	7.587*** (0.078)
Group	Eastern	Central	Western	SOEs	FFs	PFs
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,424,064	75,086	62,085	407,047	254,504	899,684
Adjusted R ²	0.629	0.763	0.715	0.651	0.721	0.612
Panel C: Dual-Regime Firms						
Treat _{it}	0.071*** (0.021)	-0.021 (0.044)	-0.035 (0.115)	0.147*** (0.034)	0.066** (0.034)	0.017 (0.024)
Constant	7.711*** (0.020)	8.139*** (0.039)	7.439*** (0.110)	7.823*** (0.033)	7.598*** (0.031)	7.864*** (0.022)
Group	Eastern	Central	Western	SOEs	FFs	PFs
Firm-Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9,251,945	370,698	168,226	2,058,914	5,037,805	2,694,150
Adjusted R ²	0.622	0.738	0.737	0.619	0.631	0.642

Notes: This table presents the heterogeneous effects of the ENE on exports by firms engaged exclusively in processing exports, general exports, or both, across different regions and ownership types. Each panel column represents a separate regression estimate based on equation (18), with the dependent variable being the logarithm of exports. Panels A, B, and C focus on firms involved solely in processing exports, general exports, and mixed exports, respectively. Columns (1)-(3) in each panel present results for firms in Eastern, Central, and Western China, while Columns (4)-(6) provide estimates for state-owned enterprises (SOEs), foreign firms, and private firms. Missing results in Columns (3) and (4) of Panel A are due to the limited availability of observations. Robust standard errors are clustered at the firm level and are reported in parentheses. Statistical significance is denoted by *** at the 1% level, ** at the 5% level, and * at the 10% level.

Table B13: Average Effects of Expressway Proximity on Export Prices

	ln(price)			
	(1)	(2)	(3)	(4)
Panel A: Baseline Treatment Definition (20 km Radius)				
Treat _{it}	-0.043*** (0.009)	-0.055*** (0.009)	0.074*** (0.025)	0.042 (0.039)
Constant	6.490*** (0.007)	6.487*** (0.008)	6.567*** (0.021)	6.508*** (0.036)
District		Eastern	Central	Western
Firm-Product FE	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes
Observations	35,786,072	32,836,500	1,689,358	1,260,214
Adjusted R ²	0.897	0.894	0.926	0.902
Panel B: Alternative Treatment Definition—County Intersection				
Treat _{it} (County)	-0.038*** (0.008)	-0.049*** (0.008)	0.096*** (0.029)	0.011 (0.043)
Constant	6.479*** (0.005)	6.474*** (0.005)	6.570*** (0.018)	6.540*** (0.027)
District		Eastern	Central	Western
Firm-Product FE	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes
Observations	35,930,382	32,836,500	1,689,358	1,260,214
Adjusted R ²	0.897	0.894	0.926	0.902
Panel C: Continuous Exposure Specification				
ln(Distance _{it})	0.018*** (0.004)	0.023*** (0.005)	-0.022* (0.011)	-0.042*** (0.015)
Constant	6.294*** (0.039)	6.240*** (0.043)	6.825*** (0.101)	6.908*** (0.127)
District		Eastern	Central	Western
Firm-Product FE	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes
Observations	35,930,382	32,836,500	1,689,358	1,260,214
Adjusted R ²	0.897	0.894	0.926	0.902

Notes: This table reports the average effects of the ENE on firm export prices under alternative treatment definitions. Panel A defines treatment using a 20km proximity threshold to the expressway network. Column (1) includes the full sample, while Columns (2)–(4) report results separately for Eastern, Central, and Western China. Panel B defines treatment at the county level based on whether the county is intersected by the expressway network in a given year. Panel C uses a continuous measure of exposure, defined as the logarithm of the distance from the firm to the expressway network, and reports the implied elasticity of exports with respect to distance. All specifications include firm–product and country–year fixed effects. Standard errors are clustered at the firm level and reported in parentheses. *** Statistically significant at the 1% level; ** at the 5% level; and * at the 10% level.

Table B14: Effects of Expressway Proximity on Extensive Margins of Trade by Ownership

	Entry			Exit		
	(1)	(2)	(3)	(4)	(5)	(6)
Treat _{it}	0.024* (0.014)	-0.025*** (0.006)	0.027*** (0.007)	0.027* (0.016)	0.008* (0.005)	-0.012*** (0.005)
Constant	0.162*** (0.013)	0.176*** (0.005)	0.198*** (0.006)	0.123*** (0.014)	0.090*** (0.004)	0.118*** (0.004)
Group	SOEs	FFs	PFs	SOEs	FFs	PFs
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	68,335	772,679	689,900	65,351	720,929	583,163
Adjusted R ²	0.350	0.245	0.175	0.157	0.153	0.170
	ln(Number of Product)			ln(Number of Destination Country)		
	(7)	(8)	(9)	(10)	(11)	(12)
Treat _{it}	0.005 (0.040)	0.017 (0.014)	0.036*** (0.012)	0.011 (0.042)	0.033** (0.015)	0.037*** (0.014)
Constant	1.760*** (0.037)	1.315*** (0.013)	1.211*** (0.011)	1.637*** (0.038)	1.229*** (0.014)	1.408*** (0.013)
Group	SOEs	FFs	PFs	SOEs	FFs	PFs
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	69,621	784,973	690,760	69,621	784,973	690,760
Adjusted R ²	0.758	0.708	0.708	0.734	0.759	0.712

Notes: This table explores the effects of the ENE on firms' entries and exits by using the firm-year level data in Sample A to estimate equation (18). The dependent variables are the dummy of firms' entries or exits. The regressions also control for firm and county-year fixed effects, and the robust standard errors are clustered at the firm level. *** Statistically significant at the 1% level; ** 5% level; and * 10% level.

Appendix C: Data

Processing Chinese Industry Firm Data — To ensure data reliability and eliminate reporting errors, we clean the CASIF database by excluding entries that violated generally accepted accounting principles (GAAP), as outlined by [Cai and Liu \(2009\)](#) and [Wang \(2023\)](#). Specifically, the following criteria are applied: (1) Firm identification numbers must be unique and non-missing; (2) The establishment year must be between 1800 and the current year; (3) Sales revenue must exceed 5 million CNY before 2011 and 20 million CNY from 2011 onward; (4) Main financial variables (e.g., total assets, annual payroll, intermediate inputs, net capital balance, gross output) must be greater than zero; (5) The number of employees must be at least 10; (6) Fixed and current assets must be less than or equal to total assets; (7) Current-year depreciation cannot exceed accumulated depreciation; (8) Income tax cannot exceed total assets or sales profit. Applying these criteria led to the removal of 8.6% of the observations.

Some crucial variables in the CASIF dataset (e.g., value-added tax, depreciation, annual payroll, value-added) are missing in specific years, especially the data after 2008, which affects total factor productivity (TFP) calculations. To address this, we employ imputation based on accounting standards: (1) Value-Added Tax is calculated by “Output tax - Input tax”; (2) Set the rental rate of capital as 10% ([Hsieh and Klenow, 2009](#)) and fill depreciation of fixed assets by using fixed assets multiplied by 10%; (3) Calculate the average ratio of each firm’s annual payroll to its sales revenue from 2000 to 2007, and use this ratio to estimate annual payroll for 2009 and 2010 by multiplying it by the corresponding sales revenue; (4) Value-added is imputed using two accounting standards: “value-added = Total output - Intermediate inputs + Value-added tax” and “Value-added = Sales revenue - Beginning inventory + Ending inventory - Intermediate inputs + Value-added tax”; (5) Intermediate inputs variable is further imputed based on the formula “Intermediate inputs = Total output * Main business cost/Main business income - Annual payroll - Depreciation of fixed assets.” These methods provide a consistent framework for reconstructing missing variables, thereby improving the robustness of TFP calculations.

Appendix D: Supplementary Measurements

TFP measurement — TFP estimates are commonly derived from production function estimations, where total factor productivity corresponds to the residuals obtained after regressing firm-level output on input variables. A key challenge in this estimation process is that firm productivity is unobservable and often correlated with input choices, which can introduce endogeneity and bias when using ordinary least squares (OLS). To address this issue, the literature ([Akerberg et al., 2015](#); [Levinsohn and Petrin, 2003](#); [Olley and Pakes, 1996](#)) has adopted semi-parametric control function methods, which rely on input demand functions to account for unobserved productivity shocks.

Given the constraints in our dataset—particularly the absence of intermediate input data since 2008, we adopt the approach of [Akerberg et al. \(2015\)](#) as the primary method for estimating TFP. Additionally, we use the approach from [Olley and Pakes \(1996\)](#) as a robustness check.

Specifically, we use the following equation to estimate TFP:

$$\ln(\text{Value Added}_{it}) = \alpha_0 + \beta_1 * \ln(\text{labor})_{it} + \beta_2 * \ln(\text{fixed assets})_{it} + \ln(\text{investment})_{it} + \gamma_t + \varepsilon_{it} \quad (20)$$

Where $\ln(\text{Value Added}_{it})$ represents the natural logarithm of firm i 's value added in year t . Labor_{it} , Fixed assets , and investment correspond to a firm's annual employment, fixed assets value, and capital investment, respectively.

TFPR measurement — Following [Hsieh and Klenow \(2009\)](#), a firm i uses labor (L_{it}) and capital (K_{it}) to produce product in each year t at technology level A_{it} . Its production function is given by a Cobb-Douglas function, $Y_{it} = A_{it}L_{it}^{\alpha}K_{it}^{\beta}$. Its profit function can be written as $\pi_{it} = P_{it}Y_{it} - (1 + \tau_{L_{it}})wL_{it} - (1 + \tau_{K_{it}})rK_{it}$, where P_{it} is the product price, Y_{it} is the production, τ_L and τ_K are the labor and capital distortion, and w and r are the labor and capital prices. Sample B provides information on labor L_{it} , capital K_{it} , sales revenue, and annual payroll. We choose the capital interest rate as 10% by following [Hsieh and Klenow \(2009\)](#). Moreover, labor share α and capital share β are measured by the approach in [Akerberg et al. \(2015\)](#). After maximizing the firm's profit, a firm's

revenue productivity is measured by the following equation.

$$TFPR_{it} = P_{it}A_{it} = \frac{P_{it}Y_{it}}{K_{it}^{\beta}(wL_{it}^{\alpha})} \quad (21)$$